Moments of Magic Squares: Line-Sum Conditions, a Permutation Threshold, and Random Energies

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Abstract

An $n \times n$ arrangement of $1, \dots, n^2$ is semi-magic if all its row and column sums are equal. If they are, then $\sum_{i,j} s_{ij} j^m = M(n) \sum_j j^m$ and $\sum_{i,j} s_{ij} i^m = M(n) \sum_i i^m$ for every m; for real matrices the converse needs these moment identities up to $m = n-1$. For arrangements fewer suffice. Let the permutation threshold $k^*(n)$ be the least k such that every arrangement satisfying them for $m = 1, \dots, k$ is semi-magic. In terms of the column-sum deviations γ these conditions say that $\sum_j \gamma_j t^j$ is divisible by $(t-1)^{k+1}$, which links $k^*(n)$ to the Prouhet–Tarry–Escott problem and to work of Borwein, Erdélyi and Kós on integer polynomials with a zero of high order at 1. We prove $(\sqrt{2} - o(1))\sqrt{n} \le k^*(n) \le \sqrt{\pi e\,(n-1)\log(n^4/4)}$; the lower bound uses semi-magic squares with many pairs of complementary rows and the pigeonhole principle. Explicit arrangements together with short proofs give $k^*(n) = n-1$ for $3 \le n \le 13$ and $k^*(n) = n-2$ for $14 \le n \le 17$, and we prove $k^*(n) \le n-3$ for all $n \ge 18$; with computer assistance we determine $k^*(n)$ for all $n \le 40$. We also treat uniformly random arrangements: specialising the classical variance formula for quadratic assignment statistics, we give the exact mean and variance of the sum of squared line-sum deviations and of a four-term energy E_4 formed from the first moments and the two diagonals, show that E_4 divided by its mean converges in distribution to Exp(1), give the mean and variance of Loly's moment-of-inertia invariant, and count the 14,639,440 arrangements of order 4 with $E_4 = 0$, of which 7,040 are magic.

Keywords: magic squares, Prouhet–Tarry–Escott problem, polynomials with integer coefficients, permutation statistics, combinatorial central limit theorem, moment of inertia
MSC 2020: 05B20 (primary); 05A05, 11C08, 60C05, 60F05

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu appears in Chinese sources more than two millennia old, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order have been known since the seventeenth century [3, 28]. The modern literature treats them through enumeration and the geometry of the associated polytopes [32, 12, 1, 5, 4], through spectral and matrix properties [21, 26, 27, 9], and, in work of Loly and collaborators, through physical invariants such as the moment of inertia and the multipole moments [19, 30, 31].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \dots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $M(n) = n(n^2+1)/2$, and *semi-magic* if the rows and columns do.

We centre both the grid and the entries (Definition 2.1): the cell (i, j) sits at $(x_j, y_i) = (j - \frac{n-1}{2}, \frac{n-1}{2} - i)$ and carries the value $z_{ij} = s_{ij} - \frac{n^2+1}{2}$. A *test function* is a real function φ on the cells, used only through the inner product $\langle \varphi, z \rangle = n^{-2} \sum_{i,j} \varphi_{ij} z_{ij}$. The condition "line L sums to $M(n)$" says that z is orthogonal to the centred indicator of L, so an arrangement is magic if and only if z is orthogonal to a fixed $2n$-dimensional space W of test functions. The dimension count is the classical fact that the real magic squares of order $n \geq 3$ form a space of dimension $n^2 - 2n$ [37], [33, §4.6], and the squared length of the vector of line deviations, $\sum_L \delta_L^2$, is the energy of the Monte Carlo study of magic squares by Pinn and Wieczerkowski [29]. This reformulation is elementary; we use it to pose the questions treated below. The convex hull of the points (x_j, y_i, z_{ij}) is studied in the companion paper [23] and is not used here.

The permutation threshold. Replacing the line indicators by the powers x^m and y^m, $1 \leq m \leq k$, gives the *degree-k conditions* $\sum_{i,j} z_{ij} x_j^m = \sum_{i,j} z_{ij} y_i^m = 0$; equivalently, the moments of orders $1, \ldots, k$ of the vector of column-sum deviations and of the vector of row-sum deviations vanish (Lemma 3.2). They hold for every semi-magic square, and for real-valued z they imply equal row and column sums if and only if $k \geq n - 1$ (Vandermonde). For arrangements, integrality and the bounded size of the line sums lower this threshold. Let $k^*(n)$ be the least k such that every arrangement satisfying the degree-k conditions is semi-magic. A *witness* for $k^*(n) \geq K$ is an arrangement that satisfies the degree-$(K-1)$ conditions and is not semi-magic. The degree-k conditions say that the polynomial $\sum_j \gamma_j t^j$ formed from the column deviations is divisible by $(t-1)^{k+1}$, and similarly for the rows, so that the positive and negative parts of γ form a solution of a Prouhet–Tarry–Escott system [38, 7]. Our main result (Theorem 3.7) is

$$(\sqrt{2} - o(1))\sqrt{n} \;\leq\; k^*(n) \;\leq\; \sqrt{\pi e\,(n-1)\log(n^4/4)} \;\leq\; 2\sqrt{\pi e}\,\sqrt{n \log n},$$

together with the exact values $k^*(n) = n - 1$ for $3 \leq n \leq 13$ and $k^*(n) = n - 2$ for $14 \leq n \leq 17$ and the bound $k^*(n) \leq n - 3$ for all $n \geq 18$, all with short proofs, and the exact values for $n \leq 40$, which are computer-assisted (Table 2). The upper bound is a Jensen-formula argument of the type used by Borwein, Erdélyi and Kós for polynomials with bounded coefficients [8]. The lower bound needs the converse direction, namely that suitable integer vectors are column-deviation vectors of arrangements whose row sums all equal $M(n)$; we obtain it from semi-magic squares with many pairs of complementary rows, in which swaps realise prescribed column deviations, combined with the pigeonhole principle. Whether $k^*(n)$ grows like $\sqrt{n}$ or like $\sqrt{n \log n}$ remains open.

Random arrangements. For a uniformly random arrangement each $\langle \varphi, z \rangle$ is a linear permutation statistic, so the energy $E_{\text{full}} = N^{-2} \sum_L \delta_L^2$ ($N = n^2$, all $2n + 2$ lines) and the four-term energy E_4, which combines the degree-1 and the two diagonal conditions (Definition 2.4), are quadratic forms in a random permutation of the centred values. We specialise the classical variance formula for quadratic assignment statistics [22, 16] to quadratic forms with zero row sums in the centred entries (Proposition 4.3); the variance is then a combination of three invariants of the matrix of the form, and we obtain the exact means and variances of both energies for every n (Theorem 4.4). By Hoeffding's combinatorial central limit theorem, $E_4 / \mathbb{E}[E_4] \to \text{Exp}(1)$ (Corollary 4.5). The same tools give the mean and variance of Loly's moment-of-inertia invariant $I(S) = \sum s_{ij}(x_j^2 + y_i^2)$ under a uniformly random arrangement (Corollary 4.2); magic squares take exactly the mean value.

The class L_4. Exactly 14,639,440 of the 16! arrangements of order 4 satisfy the four conditions $E_4 = 0$, and 7,040 of them are magic (Proposition 5.3); the count is 0.80 times a Gaussian local-limit heuristic.

Classical and new material. Orthogonality to the line space, the Vandermonde threshold, the vanishing of the moments $\langle a(x) + b(y), z \rangle$ of a semi-magic square, Loly's invariant [19] and the means of the energies are classical or elementary, Proposition 4.3 is a special case of classical

formulas, and the single-swap formula of Remark 5.4 is the incremental evaluation used in local search. To our knowledge Theorem 3.7, the closed forms for the variances in Theorem 4.4 with the limit law, the variance in Corollary 4.2 and the count in Proposition 5.3 have not appeared before.

Organisation. Section 2 fixes notation and collects the tools; Table 1 summarises the notation. Section 3 contains the moment conditions and the permutation threshold. Section 4 treats random arrangements, Section 5 the computations for small n, and Section 6 lists open questions. The appendices contain the computational methods (Appendix A), the construction of the squares used for the lower bound (Appendix B), the derivation of the variance formulas (Appendix C) and a witness for $n = 5$ (Appendix D). The two parts are independent: Section 3 (with Appendices B and D) can be read without Sections 4 and 5 (with Appendix C) and conversely; they share only the notation of Section 2, and Appendix A describes the computations for both.

2 Notation and tools

2.1 Centred coordinates

Throughout, $n \geq 3$ and $N = n^2$. Cells are pairs $c = (i, j)$ with $0 \leq i, j \leq n-1$ (i the row from the top, j the column from the left).

Definition 2.1 (Centred coordinates)**.** To the cell (i, j) of an arrangement S assign

$$x_{ij} = x_j = j - \frac{n-1}{2}, \qquad y_{ij} = y_i = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}. \tag{1}$$

We call $z = z(S)$ the *centred entries* of S, and regard x, y and z as functions on the cells.

The function z is the vertical coordinate of the points (x_c, y_c, z_c) studied in [23]; we do not call it a height function, a term that in combinatorics refers to tilings [34]. A *line* is one of the n rows R_i, the n columns C_j, the main diagonal $D_{\mathrm{m}} = \{i = j\}$ or the anti-diagonal $D_{\mathrm{a}} = \{i + j = n-1\}$, and $\mathbf{1}_L$ denotes the indicator of a line L. Each line has n cells, and since $M(n) = n \cdot \frac{N+1}{2}$ the *deviation* of a line is

$$\delta_L(S) = \sum_{c \in L} s_c - M(n) = \sum_{c \in L} z_c. \tag{2}$$

We write ρ_i, γ_j, δ_{m}, δ_{a} for the deviations of row i, column j and the two diagonals; $\sum_i \rho_i = \sum_j \gamma_j = 0$. On the space $\mathcal{F} = \mathbb{R}^{\mathrm{cells}}$ we use $\langle f, g \rangle = \frac{1}{N} \sum_c f_c g_c$ and $\|f\| = \langle f, f \rangle^{1/2}$; $\bar{f} = \langle f, \mathbf{1} \rangle$ is the mean of f and $\tilde{f} = f - \bar{f}\mathbf{1}$ its centred version. For vectors in $\mathbb{R}^n$, such as ρ and γ, $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the usual ℓ^1 and Euclidean norms.

Lemma 2.2. *For every arrangement, $\bar{x} = \bar{y} = \bar{z} = 0$, $\|x\|^2 = \|y\|^2 = \frac{n^2-1}{12}$, $\|z\|^2 = \frac{N^2-1}{12}$, $\langle x, y \rangle = 0$, and $\langle \mathbf{1}_L, z \rangle = \delta_L / N$ for every line L.*

Proof. x, y and z take n, n and N equally spaced values centred at 0, each equally often; $\langle x, y \rangle = \frac{1}{N} \sum_j x_j \sum_i y_i = 0$; the last claim is (2). □

2.2 The line space and the energies

Let $\tilde{\mathbf{1}}_L = \mathbf{1}_L - \frac{1}{n}\mathbf{1}$ be the centred indicator of a line and $W_R = \mathrm{span}\{\tilde{\mathbf{1}}_{R_i}\}$, $W_C = \mathrm{span}\{\tilde{\mathbf{1}}_{C_j}\}$, $W_D = \mathrm{span}\{\tilde{\mathbf{1}}_{D_{\mathrm{m}}}, \tilde{\mathbf{1}}_{D_{\mathrm{a}}}\}$, $W = W_R + W_C + W_D \subset \mathbf{1}^{\perp}$. P_W denotes the orthogonal projection onto W.

Lemma 2.3 (The line space)**.** *Let $n \geq 3$. For lines L, L', $N\langle \tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'} \rangle = |L \cap L'| - 1$. Consequently:*

1. *W_R consists of the zero-mean functions of the row index and W_C of the zero-mean functions of the column index; both have dimension $n-1$, and the orthogonal projection of z onto them is $(i, j) \mapsto \rho_i/n$, resp. γ_j/n.*

2. *W_R, W_C, W_D are mutually orthogonal, and the Gram matrix of $(\tilde{\mathbf{1}}_{D_\mathrm{m}}, \tilde{\mathbf{1}}_{D_\mathrm{a}})$ is Γ_D/N with $\Gamma_D = \left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$, $\epsilon_n = 0$ for odd and -1 for even n; its eigenvalues are $n-1, n-1$ (odd n) and $n-2, n$ (even n).*

3. *$W = W_R \oplus W_C \oplus W_D$ has dimension $2n$, and $z \perp W$ if and only if S is magic.*

Proof. Each line has n of the N cells, which gives the inner products: a row meets every column and each diagonal in one cell, two distinct rows (or columns) are disjoint, and the diagonals meet in the centre cell if and only if n is odd. The projection onto the functions of the row index replaces z by its row means ρ_i/n. Finally $\langle \tilde{\mathbf{1}}_L, z\rangle = \delta_L/N$ by Lemma 2.2. □

Part 3 is equivalent to the classical fact that the real magic squares of order $n \geq 3$ (all line sums equal to a common, arbitrary value) form a space of dimension $N - 2n$ [37], [33, §4.6].

Definition 2.4 (Energies)**.** For an arrangement S with centred entries z,

$$E_{\mathrm{full}}(S) = \sum_L \langle \mathbf{1}_L, z\rangle^2 \qquad (\text{all } 2n+2 \text{ lines}), \tag{3}$$

$$E_4(S) = \langle x, z\rangle^2 + \langle y, z\rangle^2 + \langle \mathbf{1}_{D_\mathrm{m}}, z\rangle^2 + \langle \mathbf{1}_{D_\mathrm{a}}, z\rangle^2. \tag{4}$$

We call E_4 the *four-term energy.*

E_4 keeps the diagonal conditions and replaces the $2n-2$ independent row and column conditions by the two degree-1 moment conditions (Proposition 3.3); it is D_4-invariant and has a fixed number of terms, which makes it a test case for a limit law (Corollary 4.5). By Lemma 2.2,

$$N^2 E_{\mathrm{full}}(S) = \sum_L \delta_L(S)^2 = \sum_i \rho_i^2 + \sum_j \gamma_j^2 + \delta_\mathrm{m}^2 + \delta_\mathrm{a}^2, \tag{5}$$

the sum of the squared deviations of all $2n+2$ lines, a non-negative integer. This is the energy of the Monte Carlo study of Pinn and Wieczerkowski [29], who use $\sum_L \delta_L^2 = N^2 E_{\mathrm{full}}$; we follow them in calling E_{full} and E_4 energies. $E_{\mathrm{full}}(S) = 0$ if and only if S is magic, and E_{full} depends on z only through its projection onto W (Figure 1). Both energies are invariant under the dihedral group D_4 of the square, which permutes the lines and maps $\{\pm x, \pm y\}$ to itself. All n rows and n columns are kept in (3): the last row and column are redundant for the zero set, but omitting them would destroy the D_4-invariance.

Remark 2.5 (The four conditions). $E_4 = 0$ consists of four linear conditions on z, and "magic" of $2n$ independent ones; for every $n \geq 3$ the four conditions are therefore strictly weaker than the magic conditions on real-valued z. For permutations the two coincide when $n = 3$ (Proposition 5.1) but not when $n = 4$: the arrangement

$$S = \begin{pmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{pmatrix}$$

has both diagonal sums 34, row sums $(28, 38, 44, 26)$ and column sums $(37, 25, 43, 31)$, so $\gamma = (3, -9, 9, -3)$ and $\rho = (-6, 4, 10, -8)$, and $\langle x, z\rangle = \langle y, z\rangle = 0$; so $E_4(S) = 0$ while $N^2 E_{\mathrm{full}}(S) = 396$. Both orders are treated in Section 5.1.

Remark 2.6 (Linear functionals and E_{full}). By Lemma 2.3, with $\delta = (\delta_\mathrm{m}, \delta_\mathrm{a})$,

$$\|P_W z\|^2 = \frac{n\big(\|\rho\|_2^2 + \|\gamma\|_2^2\big)}{N^2} + \frac{1}{N}\,\delta^{\mathsf{T}} \Gamma_D^{-1} \delta,$$

and since the eigenvalues of Γ_D are $n-1, n-1$ for odd n and $n-2, n$ for even n, comparison with (5) gives $nE_{\mathrm{full}} \leq \|P_W z\|^2 \leq \lambda_n E_{\mathrm{full}}$, with $\lambda_n = n^2/(n-1)$ for odd n and $\lambda_n = n^2/(n-2)$ for even n. Hence $|\sum_c z_c \varphi_c| = N|\langle \varphi, P_W z\rangle| \leq N\|\varphi\|\sqrt{\lambda_n E_{\mathrm{full}}}$ for every $\varphi \in W$; for instance $x, y \in W_C \oplus W_R$, so the centre of mass $\sum_c z_c(x_c, y_c)$ is controlled by E_{full}. Nonlinear functionals are not controlled: with u_c the unit vector from the centre to cell c ($u_c = 0$ at the centre), $|\sum_c z_c u_c| = 2(2-\sqrt{2})\sqrt{5}$ for every magic square of order 3 and is positive for all 880 of order 4 (`anc/balance_check.py`).

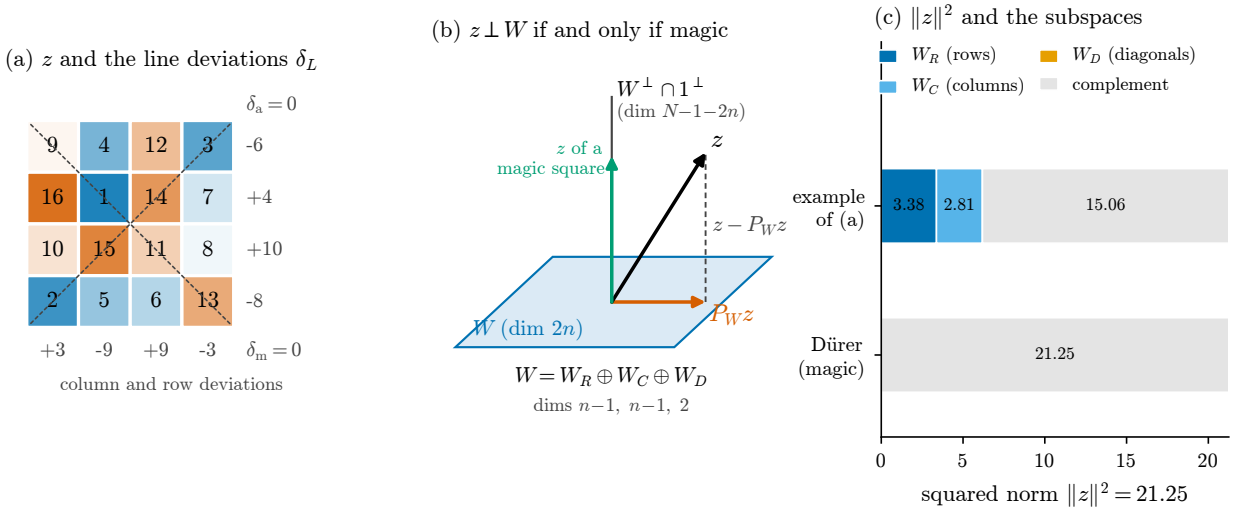


Figure 1: The line space for $n = 4$. W is the $2n$-dimensional span of the centred line indicators, and $W^\perp \cap \mathbf{1}^\perp$ (dimension $N - 1 - 2n$) is its orthogonal complement among the zero-mean functions. (a) The arrangement of Remark 2.5, shaded by z, with its row and column deviations (both diagonal deviations are 0). (b) Schematic of the orthogonal splitting $z = P_W z + (z - P_W z)$; the arrangement is magic if and only if $P_W z = 0$. (c) The squared norms $\|P_{W_R} z\|^2, \|P_{W_C} z\|^2, \|P_{W_D} z\|^2$ and that of the complement, for the arrangement of (a) $(3.38 + 2.81 + 0 + 15.06)$ and for Dürer's magic square (all of $\|z\|^2 = 21.25$ lies in the complement).

2.3 Permutation statistics

Let π be a uniformly random permutation of $\{1, \dots, N\}$ and $a, a', b \in \mathbb{R}^N$. For the linear permutation statistics $\Phi_a = \sum_i a_i b_{\pi(i)}$ one has $\mathbb{E}\Phi_a = N\bar{a}\bar{b}$ (here $\bar{a} = \frac{1}{N}\sum_i a_i$) and, by the theorem of Wald and Wolfowitz [36, 25] (polarised),

$$\operatorname{Cov}(\Phi_a, \Phi_{a'}) = \frac{1}{N-1} \sum_i (a_i - \bar{a})(a'_i - \bar{a}') \sum_j (b_j - \bar{b})^2. \tag{6}$$

For a uniformly random arrangement, $z_c = b_{\pi(c)}$ with b the centred values $b_v = v - \frac{N+1}{2}$, $1 \le v \le N$; we put

$$\sigma_n^2 = \frac{\|z\|^2}{N-1} = \frac{N+1}{12} = \frac{n^2+1}{12}. \tag{7}$$

Hoeffding's combinatorial central limit theorem [15, Thm. 4 and condition (14), p. 560] states that for real vectors $a = a^{(N)}$, $b = b^{(N)}$ the statistic $\sum_i a_i b_{\pi(i)}$ is asymptotically normal, i.e. $\big(\sum_i a_i b_{\pi(i)} - \mathbb{E}\big)/\sqrt{\operatorname{Var}}$ converges in distribution to $\mathcal{N}(0,1)$, if

$$\lim_{N\to\infty} N \frac{\max_i (a_i - \bar{a})^2 \, \max_j (b_j - \bar{b})^2}{\sum_i (a_i - \bar{a})^2 \, \sum_j (b_j - \bar{b})^2} = 0. \tag{8}$$

Motoo [24] gives a Lindeberg-type condition and Bolthausen [6] a Berry–Esseen bound.

2.4 Known counts

There are 8 magic squares of order 3, all images of the Lo Shu under D_4; 7,040 of order 4, forming the 880 D_4-orbits listed by Frénicle de Bessy [13] and recounted many times since [4]; and 275,305,224 D_4-orbits of order 5, i.e. 2,202,441,792 squares, counted by Schroeppel in 1973 and reported by Gardner [14]. For $n \ge 6$ only statistical estimates are known [29, 35].

2.5 Notation

Table 1 lists the main symbols and where they are defined.

symbol	meaning (where defined)
n, $N = n^2$, $M(n)$	order, number of cells, magic constant $n(N+1)/2$ (Definition 1.1)
$c = (i, j)$, s_c	cell in row i and column j ($0 \le i, j < n$); entry of the arrangement S
x, y, z	centred column and row coordinates, centred entries (Definition 2.1)
$\langle f, g\rangle$, $\|f\|$, $\bar{f}$, $\tilde{f}$	$N^{-1}\sum_c f_c g_c$, its norm, the mean and the centred version of f (Section 2.1)
δ_L, ρ, γ, δ	deviation of the line L; row and column deviation vectors; $(\delta_{\mathrm{m}}, \delta_{\mathrm{a}})$ (2)
W; W_R, W_C, W_D; P_W	span of the centred line indicators; its row, column and diagonal parts; orthogonal projection onto W (Lemma 2.3)
Γ_D, ϵ_n	N times the Gram matrix of the centred diagonal indicators; $\epsilon_n = 0$ (n odd), -1 (n even) (Lemma 2.3)
E_{full}, E_4	line-sum energy and four-term energy (Definition 2.4)
σ_n^2	$(n^2+1)/12$ (7)
$P_\gamma(t)$	$\sum_j \gamma_j t^j$ (Section 3.1)
$\Lambda_{n,k}$	$\{\gamma \in \mathbb{Z}^n : (t-1)^{k+1} \mid P_\gamma\}$ (Section 3.2)
$k^*(n)$	permutation threshold (Definition 3.6)
$B_r(g)$	binomial moment $\sum_j \binom{j}{r} g_j$ of a vector g (Section 3.3)
H	$\lfloor n^3/4 \rfloor$ (Section 3.3)
r_q, F_q	number of pairs of complementary rows covering the q-th column pair; set of subset sums (Proposition 3.13)
T_φ, τ_ℓ	$\langle \varphi, z\rangle$ for a random arrangement; standard deviation of $T_\ell = T_{\varphi_\ell}$ (Section 4)
Γ, C_n	Gram matrix of the centred test functions of an energy; correlation matrix of the four statistics of E_4 (Proposition 4.1, Corollary 4.5)
A, h_c	matrix of an energy as a quadratic form in z; $h_c = A_{cc}$ (Section 4.2, Appendix C)
L_n, $U_1, \dots, U_4$, Σ_U	arrangements with $E_4 = 0$; the four additive statistics defining L_n; their covariance matrix (Section 5.1)

Table 1: Notation.

3 Moment conditions and the permutation threshold

3.1 Moment conditions

For a centred $f \in \mathcal{F}$ let $\gamma_j(f) = \sum_i f_{ij}$ and $\rho_i(f) = \sum_j f_{ij}$ be its column and row sums; for $f = z(S)$ these are the deviations γ, ρ of S. For $\gamma \in \mathbb{R}^n$ put $P_\gamma(t) = \sum_{j=0}^{n-1} \gamma_j t^j$.

Definition 3.1 (Degree-k conditions)**.** A centred $f \in \mathcal{F}$ satisfies the *degree-k conditions* ($k \ge 1$) if $\langle x^m, f\rangle = \langle y^m, f\rangle = 0$ for $1 \le m \le k$. For $f = z(S)$ these read $\sum_{i,j} z_{ij} x_j^m = \sum_{i,j} z_{ij} y_i^m = 0$.

Lemma 3.2 (Polynomial form)**.** *If φ depends only on the column index, $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; similarly for rows. For $\gamma \in \mathbb{R}^n$ and $k \ge 0$ the following are equivalent: (i) $\sum_j x_j^m \gamma_j = 0$ for $0 \le m \le k$; (ii) $\sum_j \binom{j}{r}\gamma_j = 0$ for $0 \le r \le k$; (iii) $(t-1)^{k+1}$ divides $P_\gamma(t)$. Hence a centred f satisfies the degree-k conditions if and only if $(t-1)^{k+1}$ divides both $P_{\gamma(f)}$ and $P_{\rho(f)}$.*

Proof. The first statement is $\sum_{i,j}\varphi(j) f_{ij} = \sum_j \varphi(j)\gamma_j(f)$. The families $(x_j^m)_{m\le k}$ and $\left(\binom{j}{r}\right)_{r\le k}$ both span the restrictions to $\{0, \dots, n-1\}$ of the polynomials of degree at most k in j, so (i)$\Leftrightarrow$(ii); and $P_\gamma(t) = \sum_r \left(\sum_j \binom{j}{r}\gamma_j\right)(t-1)^r$ is the Taylor expansion at $t = 1$, so (ii)$\Leftrightarrow$(iii). The case $m = 0$ of (i) holds for every centred f. □

Proposition 3.3 (Moment conditions)**.** *Let $n \ge 3$ and let $f \in \mathcal{F}$ be centred.*

(a) If $\gamma(f) = \rho(f) = 0$, then $\langle a(x) + b(y), f\rangle = 0$ for all functions a, b on $\{x_0, \dots, x_{n-1}\}$; in particular f satisfies the degree-k conditions for every k. For $f = z(S)$ with S magic, also $\langle a(x) + b(y) + c_1 \mathbf{1}_{D_{\mathrm{m}}} + c_2 \mathbf{1}_{D_{\mathrm{a}}}, z\rangle = 0$.

(b) If $k \geq n-1$, the degree-k conditions imply $\gamma(f) = \rho(f) = 0$.

(c) If $1 \leq k \leq n-2$, the function $f_{ij} = \frac{1}{n}(-1)^j \binom{k+1}{j}$ $(0 \leq j \leq k+1)$, $f_{ij} = 0$ $(j > k+1)$, satisfies the degree-k conditions but not the degree-$(k+1)$ conditions, and $\gamma(f) \neq 0$.

Proof. (a) is the first statement of Lemma 3.2. (b) A nonzero polynomial of degree at most $n-1$ is not divisible by $(t-1)^n$. (c) Here $\rho(f) = 0$ and $P_{\gamma(f)}(t) = (1-t)^{k+1}$. □

Part (a) contains the vanishing of the centre of mass $\sum_c z_c(x_c, y_c)$ and of all axial moments $\sum_c z_c x_c^m$, $\sum_c z_c y_c^m$ of a semi-magic square, in the spirit of the multipole analysis of a magic cube by Rogers and Loly [31]. Parts (b) and (c) are the invertibility of a Vandermonde matrix. The mixed moments are not determined by the line sums:

Remark 3.4 (Mixed moments). The magic square with rows $(1, 6, 11, 16)$, $(8, 15, 2, 9)$, $(12, 3, 14, 5)$, $(13, 10, 7, 4)$ has mixed moment $\sum_c z_c x_c y_c = 60$ (Loly's product of inertia). Among the 880 magic squares of order 4 up to D_4, the mixed moment $\sum_c z_c x_c y_c$ is nonzero for 824 and takes 48 distinct absolute values between 0 and 60, and the mixed moments $\sum_c z_c x_c^2 y_c$, $\sum_c z_c x_c y_c^2$ are nonzero for 623 and 773 of them. For $n = 3$ the mixed moment $\sum_c z_c x_c y_c$ vanishes for all eight magic squares, because $s_c + s_{-c} = 10$ for centrally opposite cells, so z is odd under $(x, y) \mapsto (-x, -y)$ while xy is even.

Corollary 3.5 (Loly's invariant [19])**.** *Let $I(S) = \sum_{i,j} s_{ij}(x_j^2 + y_i^2)$ be the moment of inertia about the centre of the masses s_{ij} placed at (x_j, y_i). For every arrangement,*

$$I(S) = N\,\langle x^2 + y^2, z\rangle + \frac{n^2(n^4-1)}{12} = \sum_j x_j^2 \gamma_j + \sum_i y_i^2 \rho_i + \frac{n^2(n^4-1)}{12}. \tag{9}$$

Hence $I(S) = n^2(n^4-1)/12$ if and only if the single condition $\langle x^2 + y^2, z\rangle = 0$ holds; it is implied by the degree-2 conditions, hence by the semi-magic property.

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$ and use Lemma 2.2 and Lemma 3.2: $\frac{N+1}{2}\sum_{i,j}(x_j^2 + y_i^2) = \frac{N+1}{2} \cdot \frac{n^2(n^2-1)}{6}$. □

Loly's derivation uses the row and column sums in the same way [19]. The single condition is much weaker than the degree-2 conditions: for $n = 3$, 24,192 of the 9! arrangements have $I = 60$, while 72 (the semi-magic squares, Remark 5.2) satisfy the degree-2 conditions. The energy E_4 of (4) combines the degree-1 conditions with the two diagonal conditions: $E_4 = \langle x, z\rangle^2 + \langle y, z\rangle^2 + (\delta_{\mathrm{m}}^2 + \delta_{\mathrm{a}}^2)/N^2$.

3.2 The permutation threshold

By Proposition 3.3, for real-valued f the degree-k conditions force equal row and column sums exactly when $k \geq n-1$. For arrangements the deviations are integers of bounded size, and the threshold drops.

Definition 3.6. For $n \geq 3$, $k^*(n)$ is the least $k \geq 1$ such that every arrangement of order n satisfying the degree-k conditions is semi-magic. A *witness* for $k^*(n) \geq K$ is an arrangement that satisfies the degree-$(K-1)$ conditions and is not semi-magic.

By Proposition 3.3(b), $k^*(n) \leq n-1$. By Lemma 3.2, an arrangement satisfies the degree-k conditions if and only if γ and ρ lie in

$$\Lambda_{n,k} = \{\gamma \in \mathbb{Z}^n : (t-1)^{k+1} \mid P_\gamma(t)\} = (t-1)^{k+1}\,\mathbb{Z}[t]_{<n-k-1}$$

(identifying γ with P_γ; the quotient of an integer polynomial by the monic polynomial $(t-1)^{k+1}$ has integer coefficients), a lattice of rank $n-k-1$. Writing $\gamma = \gamma^+ - \gamma^-$ with $\gamma^\pm \geq 0$, the condition $\gamma \in \Lambda_{n,k}$ says that the multisets in which j occurs γ_j^+, resp. γ_j^-, times have equal power sums of

degrees $0, \dots, k$: a Prouhet–Tarry–Escott system [38, 7]. Small vectors in $\Lambda_{n,k}$ are classical: the Prouhet–Thue–Morse vector $\big((-1)^{s_2(j)}\big)_{0 \le j < 2^{k+1}}$ (s_2 the binary digit sum) has entries ± 1 and lies in $\Lambda_{2^{k+1},k}$ [38, 2], and Borwein, Erdélyi and Kós construct polynomials of degree n with coefficients in $\{-1, 0, 1\}$ vanishing to order $\ge c\sqrt{n/\log n}$ at 1 [8, Thm. 2.7]. The question is therefore how large k can be for vectors that are column-deviation vectors of arrangements.

Theorem 3.7 (The permutation threshold)**.**

(a) For all $n \ge 3$, $k^*(n) \le \Big\lfloor \sqrt{\pi e\,(n-1)\log(n^4/4)} \Big\rfloor \le 2\sqrt{\pi e}\,\sqrt{n \log n}$.

(b) $\displaystyle\liminf_{n\to\infty} \frac{k^*(n)}{\sqrt{n}} \ge \sqrt{2}$.

(c) $k^*(n) = n-1$ for $3 \le n \le 13$, $k^*(n) = n-2$ for $14 \le n \le 17$, and $k^*(n) \le n-3$ for every $n \ge 18$. The values for $14 \le n \le 40$ are those of Table 2.

The constant $2\sqrt{\pi e}$ is about 5.84. Part (a) gives $k^*(n) \le n-3$ from $n = 166$ on and improves on it from $n = 169$; it is of asymptotic interest only, since part (c) gives $k^*(n) \le n-3$ for all $n \ge 18$. In part (c) the lower bounds rest on explicit witness arrangements, checkable in exact arithmetic; $k^*(n) \le n-1$, $k^*(n) \le n-2$ ($n \ge 14$) and $k^*(n) \le n-3$ ($n \ge 18$) are proved by hand; only the upper bounds of Table 2 for $18 \le n \le 40$ are computer-assisted (Section 3.5).

3.3 Proof of the upper bound

Lemma 3.8 (Majorisation bound for column deviations)**.** *Let γ be the column-deviation vector of an arrangement. For every set J of p columns,*

$$\Big|\sum_{j\in J} \gamma_j\Big| \le \frac{pn(N-pn)}{2}. \tag{10}$$

In particular $|\gamma_j| \le n^2(n-1)/2$, $\|\gamma\|_1 \le N^2/4$ and $\|\gamma\|_2^2 \le n^5(n^2-1)/12$. The same holds for ρ.

Proof. The pn cells of J carry distinct values, whose sum is at most that of the pn largest values, $pn(2N - pn + 1)/2$; subtracting $pM(n) = pn(N+1)/2$ gives the upper bound, and the lower bound is symmetric. With p the number of positive γ_j, $\|\gamma\|_1 = 2\sum_{\gamma_j>0}\gamma_j \le pn(N-pn) \le N^2/4$. Finally, γ is majorised by the vector v with $v_p = n(N-(2p-1)n)/2$, $1 \le p \le n$ (equal totals, $\sum_p v_p = 0 = \sum_j \gamma_j$, and (10) bounds the partial sums of γ sorted in decreasing order by $\sum_{p\le P} v_p = Pn(N-Pn)/2$); $\|v\|_2^2 = n^5(n^2-1)/12$, and $\|\cdot\|_2^2$ is Schur-convex. □

Lemma 3.9 (Order of vanishing of an integer polynomial at 1)**.** *Let $\gamma \in \mathbb{Z}^n \setminus \{0\}$ with $(t-1)^K \mid P_\gamma$ for some $K \ge 1$. Then $K^2 \le \pi e\,(n-1)\log\|\gamma\|_1$.*

Proof. Write $P = P_\gamma$ and $L = \|\gamma\|_1$; note $K \le \deg P \le n-1$. The Mahler measure $\mathcal{M}(P) = |a|\prod_\alpha \max(1,|\alpha|)$ of a nonzero $P \in \mathbb{Z}[t]$, with a its leading coefficient and the product over its roots α, is at least 1, because $|a| \ge 1$, and Jensen's formula gives $\log \mathcal{M}(P) = \int_{-1/2}^{1/2} \log|P(e^{2\pi i\theta})|\,d\theta$. A root at 0 (when $\gamma_0 = 0$) contributes $\max(1,0) = 1$, so no condition on γ_0 is needed. Hence $\int_{-1/2}^{1/2} \log|P(e^{2\pi i\theta})|\,d\theta \ge 0$. On $|z| = 1$ we have $|P(z)| \le L$. Since $P^{(i)}(1) = 0$ for $i < K$, Taylor's formula with integral remainder along the segment $[1, z]$, which lies in the closed unit disc when $|z| = 1$, gives

$$|P(z)| \le \frac{|z-1|^K}{K!}\max_{|w|\le 1}|P^{(K)}(w)| \le L\binom{n-1}{K}|z-1|^K \le L\Big(\frac{e(n-1)|z-1|}{K}\Big)^K,$$

because $|P^{(K)}(w)| \le \sum_j |\gamma_j|\,j(j-1)\cdots(j-K+1)$ for $|w| \le 1$. With $|e^{2\pi i\theta}-1| \le 2\pi|\theta|$ and $\beta = 2\pi e(n-1)/K \ge 2\pi e$ we get $\log|P(e^{2\pi i\theta})| \le \log L + K\log\min(1, \beta|\theta|)$, and $\int_{-1/2}^{1/2}\log\min(1,\beta|\theta|)\,d\theta = -2/\beta$. Hence $0 \le \log L - 2K/\beta = \log L - K^2/(\pi e(n-1))$. □

Proof of Theorem 3.7(a). Let S satisfy the degree-k conditions with $(k+1)^2 > \pi e(n-1)\log(n^4/4)$. If $\gamma \neq 0$, then $(t-1)^{k+1} \mid P_\gamma$ (Lemma 3.2) and $\|\gamma\|_1 \le n^4/4$ (Lemma 3.8), contradicting Lemma 3.9. So $\gamma = 0$, and likewise $\rho = 0$. The second inequality follows from $(n-1)\log(n^4/4) < 4n\log n$. □

The order $\sqrt{n\log n}$ also follows from [8, Thm. 2.1], applied to $P_\gamma(t)/(t^r \max_j |\gamma_j|)$ with r the index of the first nonzero γ_j; that theorem bounds the number of zeros at 1 of $\sum_{j\le n} a_j t^j$ with $|a_j| \le 1$ by $c\sqrt{n(1-\log|a_0|)}$, with an unspecified absolute constant. Lemma 3.9 is a self-contained version for integer coefficients.

The hypothesis $\|\gamma\|_1 \le n^4/4$ alone cannot give an upper bound of smaller order than $\sqrt{n}$. For a vector $g = (g_0, \dots, g_{n-1})$ write $B_r(g) = \sum_j \binom{j}{r} g_j$ for its binomial moments, and let $H = \lfloor n^3/4 \rfloor$. The $(H+1)^n$ vectors $g \in \{0, \dots, H\}^n$ have $B_r(g) \in \big[0, H\binom{n}{r+1}\big]$. If $(H+1)^n > \prod_{r<K}\big(H\binom{n}{r+1}+1\big)$, two of them, $g \neq g'$, share $B_0, \dots, B_{K-1}$; then $\gamma = g - g' \neq 0$, $\|\gamma\|_1 \le nH \le n^4/4$, and $(t-1)^K \mid P_\gamma$ by Lemma 3.2. Since $\prod_{r<K}\big(H\binom{n}{r+1}+1\big) \le (2H)^K n^{K(K+1)/2}$, for $K = \lfloor 2\sqrt{n} \rfloor$ the logarithm of the right side is $2n\log n + O(\sqrt{n}\log n)$, while that of the left side is at least $n\log H = 3n\log n - O(n)$. So for all large n there are such vectors with $K = \lfloor 2\sqrt{n} \rfloor$.[1]

3.4 Proof of the lower bound

A lower bound $k^*(n) \ge K$ requires a witness, i.e. an arrangement that satisfies the degree-$(K-1)$ conditions without being semi-magic. We keep all row sums equal to $M(n)$ and control the column deviations by swaps.

Definition 3.10. Rows $i \neq i'$ of an arrangement are *complementary on* a set J of columns if $s_{ij} + s_{i'j} = N+1$ for all $j \in J$.

Lemma 3.11 (Swaps in complementary rows)**.** *Let S be semi-magic, let rows i, i' be complementary on J, let $j, j' \in J$ and $d = s_{ij'} - s_{ij}$. Exchanging the entries of row i in columns j, j' changes the column deviations by $d(e_j - e_{j'})$, exchanging those of row i' changes them by $-d(e_j - e_{j'})$, and neither changes a row sum. Exchanges involving pairwise disjoint sets of cells can be combined, and their effects add.*

Proof. $s_{i'j} - s_{i'j'} = (N+1-s_{ij}) - (N+1-s_{ij'}) = d$. □

Lemma 3.12 (Squares with many complementary rows)**.**

1. *If $n = 2h$ with h even, or with h odd and $h \ge 7$, there is a semi-magic square of order n in which rows r and $h+r$ are complementary on all columns, for $0 \le r < h$.*
2. *If $n \ge 9$ is odd, there is a semi-magic square of order n in which rows ℓ and $n-1-\ell$ are complementary on the columns $\ell+1, \dots, n-2-\ell$, for $0 \le \ell \le (n-9)/2$.*

Part (1) does not cover $n = 6, 10$; this is irrelevant for Theorem 3.7(b), which concerns large n. The proof is a direct construction, given in Appendix B.

Proposition 3.13 (Pigeonhole criterion)**.** *Let S be a square of Lemma 3.12. Pair the columns as (j_q, j_q+1), where $j_q = 2q$ for $q < Q = n/2$ (n even) and $j_q = 2q+1$ for $q < Q = (n-3)/2$ (n odd), and let r_q be the number of pairs of complementary rows of Lemma 3.12 that are complementary on both columns of the q-th pair: $r_q = n/2$ for even n, and $r_q = \min(2q,\, n-4-2q,\, (n-9)/2)+1$ for odd n. For odd n the q-th pair consists of the columns $2q+1, 2q+2$, and the complementary rows $\ell, n-1-\ell$ of Lemma 3.12(2) cover both columns if and only if $\ell \le 2q$ and $\ell \le n-4-2q$; with $0 \le \ell \le (n-9)/2$ this gives the formula for r_q. If $K \ge 2$ and*

$$\prod_{q<Q}\Big(\Big\lceil \frac{r_q}{2} \Big\rceil + 1\Big) \;>\; \prod_{s=0}^{K-2}\Big((N-1)\sum_{q<Q} r_q \binom{q}{s} + 1\Big), \tag{11}$$

[1] Estimating the product as in the proof of Theorem 3.7(b) below, with $\binom{n}{r+1} \le n^{r+1}/(r+1)!$, gives such vectors with $K = (\sqrt{12} - o(1))\sqrt{n}$.

then there is an arrangement with all row sums $M(n)$ and $\gamma \neq 0$ that satisfies the degree-$(K-1)$ conditions; hence $k^*(n) \geq K$.

Proof. For each q, Lemma 3.11 gives r_q nonzero integers $d_{q,1}, \dots, d_{q,r_q}$ with $|d_{q,i}| \leq N-1$, one for each pair of complementary rows covering the q-th column pair. At least $\lceil r_q/2 \rceil$ of them have the same sign, and their partial sums, together with 0, form a set F_q of $\lceil r_q/2 \rceil + 1$ distinct subset sums lying in an interval of length at most $r_q(N-1)$. For $g \in \prod_q F_q$ the binomial moments $B_s(g) = \sum_q g_q \binom{q}{s}$, $0 \leq s \leq K-2$, are integers in intervals of lengths at most $(N-1)\sum_q r_q\binom{q}{s}$, so by (11) there are $g \neq g'$ with $B_s(g) = B_s(g')$ for all $s \leq K-2$. Write $g_q = \sum_{i \in I_q} d_{q,i}$ and $g'_q = \sum_{i \in I'_q} d_{q,i}$. For every q, perform the exchange of Lemma 3.11 in the first row of each pair in $I_q \setminus I'_q$ and in the second row of each pair in $I'_q \setminus I_q$; all these exchanges involve disjoint cells. The result has all row sums $M(n)$ and $\gamma = \sum_q \zeta_q (e_{j_q} - e_{j_q+1})$ with $\zeta = g - g' \neq 0$. Then $P_\gamma(t) = t^{j_0}(1-t)P_\zeta(t^2)$ with $P_\zeta(u) = \sum_q \zeta_q u^q \neq 0$. Since $B_s(\zeta) = 0$ for $s \leq K-2$, Lemma 3.2 gives $(u-1)^{K-1} \mid P_\zeta(u)$, so $(t^2-1)^{K-1} \mid P_\zeta(t^2)$ and $(t-1)^K \mid P_\gamma$. With $\rho = 0$, the arrangement satisfies the degree-$(K-1)$ conditions, and $\gamma \neq 0$. □

Proof of Theorem 3.7(b). Fix $\alpha < \sqrt{2}$ and let $K - 1 = k = \lfloor \alpha\sqrt{n} \rfloor$; we show that (11) holds for all large n covered by Lemma 3.12 (every $n \geq 14$ is covered).

Right side. $\sum_{q<Q} \binom{q}{s} = \binom{Q}{s+1} \leq \binom{n}{s+1}$, $r_q \leq n$ and $N \leq n^2$, so each factor is at most $n^4\binom{n}{s+1} \leq n^4\, n^{s+1}/(s+1)!$. From $\log s! \geq s\log s - s$ and $\sum_{s \leq k} s\log s \geq \int_0^k x \log x\, dx = \frac{k^2}{2}\log k - \frac{k^2}{4}$,

$$\sum_{s=1}^{k} \log s! \;\geq\; \frac{k^2}{2}\log k - \frac{3}{4}k^2 - \frac{k}{2},$$

hence $\sum_{s=1}^{k}(s\log n - \log s!) \leq \frac{k^2}{2}\log\frac{n}{k} + O(k^2 + k\log n)$ and

$$\log(\text{right side}) \leq 4k\log n + \sum_{s=1}^{k}\big(s\log n - \log s!\big) \leq \frac{k^2}{2}\log\frac{n}{k} + O(k^2 + k\log n) = \frac{\alpha^2}{4}\, n\log n + O(n).$$

Left side. For even n it equals $(\lceil n/4 \rceil + 1)^{n/2}$, with logarithm $\frac{n}{2}\log n - O(n)$. For odd n, with $Q = (n-3)/2$, $r_q = \min(2q+1,\, 2(Q-q),\, Q-2) \geq \min(q+1,\, Q-q)$ once $Q \geq 5$ ($n \geq 13$), so $\lceil r_q/2 \rceil + 1 \geq \min(q+1, Q-q)/2$. Hence

$$\log(\text{left side}) \;\geq\; \sum_{q<Q} \log\min(q+1, Q-q) - Q\log 2 \;\geq\; 2\log\big(\lfloor Q/2 \rfloor!\big) - Q\log 2 = \frac{n}{2}\log n - O(n).$$

Since $\alpha^2/4 < 1/2$, (11) holds for large n, and $k^*(n) \geq \lfloor \alpha\sqrt{n} \rfloor + 1$. □

The convergence in (b) is slow: (11) itself gives only $k^*(n) \geq 2$ at $n = 14$ and $k^*(n) \geq 7$ at $n = 100$ (Figure 2). Thus

$$(\sqrt{2} - o(1))\sqrt{n} \leq k^*(n) \leq (2\sqrt{\pi e} + o(1))\sqrt{n\log n}.$$

The situation resembles that of polynomials with coefficients in $\{-1, 0, 1\}$ in [8]: in both settings the upper bound comes from Jensen's formula and the lower construction from the pigeonhole principle, and they differ by a factor $\sqrt{\log n}$.

3.5 Exact values

Proof of Theorem 3.7(c). $3 \leq n \leq 13$. Here $k^*(n) \leq n-1$, and the lower bound needs a witness for $k^*(n) \geq n-1$, i.e. an arrangement satisfying the degree-$(n-2)$ conditions that is not semi-magic. For $n = 3$ and $n = 4$,

$$\begin{pmatrix} 1 & 3 & 9 \\ 8 & 6 & 5 \\ 7 & 4 & 2 \end{pmatrix} \quad \gamma = (1, -2, 1),\ \rho = (-2, 4, -2); \qquad \begin{pmatrix} 14 & 2 & 5 & 13 \\ 8 & 15 & 7 & 4 \\ 12 & 3 & 9 & 10 \\ 1 & 11 & 16 & 6 \end{pmatrix} \quad \gamma = (1, -3, 3, -1),\ \rho = 0.$$

In both cases $P_\gamma = (1-t)^{n-1}$ and $P_\rho \in \{0, -2(1-t)^2\}$, so the degree-$(n-2)$ conditions hold. For $5 \le n \le 13$ we found arrangements with $\rho = 0$ and $\gamma_j = (-1)^j \binom{n-1}{j}$, i.e. $P_\gamma = (1-t)^{n-1}$ (Appendix D).

The lattices $\Lambda_{n,n-2}$ and $\Lambda_{n,n-3}$. $\Lambda_{n,n-2}$ has rank 1 and is spanned by $v = \big((-1)^j \binom{n-1}{j}\big)_j$. If $\gamma = av \neq 0$ with $a \in \mathbb{Z}$, then $\max_j |\gamma_j| \ge \binom{n-1}{\lfloor (n-1)/2 \rfloor}$, which exceeds $n^2(n-1)/2$ for $n \ge 14$, contradicting Lemma 3.8. (At $n = 14$, $\binom{13}{6} = 1716 > 1274$; passing from n to $n+1$ multiplies the left side by 2 for even n and by $2n/(n+1) \ge 15/8$ for odd $n \ge 15$, and the right side by $(n+1)^2/(n(n-1))$, which decreases in n and equals $225/182 < 15/8$ at $n = 14$.) So $k^*(n) \le n-2$ for $n \ge 14$; at $n = 13$ the bound fails ($\binom{12}{6} = 924 < 1014$), and the witness has central entry 924.

$\Lambda_{n,n-3}$ has rank 2: $P_\gamma = (t-1)^{n-2}(a+bt)$ with $a, b \in \mathbb{Z}$. At $t = -1$, $2^{n-2}|a-b| = |P_\gamma(-1)| \le \|\gamma\|_1 \le N^2/4 = n^4/4$ by Lemma 3.8; since $n^4 < 2^n$ for $n \ge 17$, $a = b$ and $P_\gamma = a(t-1)^{n-2}(t+1)$. By Parseval, $\|\gamma\|_2^2 = a^2 \sum_j \big(\binom{n-2}{j} - \binom{n-2}{j-1}\big)^2 = 2a^2 \binom{2n-4}{n-2}/(n-1)$ (twice a Catalan number). At $n = 18$ this factor is 70,715,340 $>$ 50,860,872 $= 18^5(18^2-1)/12$. From n to $n+1$ the left side grows by the factor $2(2n-3)/n \ge 11/3$ and the right side by $(n+1)^4(n+2)/(n^4(n-1)) < 3/2$. So for $n \ge 18$ the bound $\|\gamma\|_2^2 \le n^5(n^2-1)/12$ of Lemma 3.8 forces $a = 0$, and $k^*(n) \le n-3$.

Upper bounds for $18 \le n \le 40$. (For $n \le 17$ they were proved above.) If no nonzero vector of $\Lambda_{n,k}$ satisfies (10) for every p, then $k^*(n) \le k$. By Lemma 3.8 only vectors with $\|\gamma\|_2^2 \le n^5(n^2-1)/12$ need be examined; we enumerated them exactly with two independent implementations (Appendix A). At $k = k^*(n)$ the ball contains no nonzero vector for 16 of the 23 values of n, since the Gram–Schmidt norms of the reduced basis already exceed the radius. For $n = 20, 22, 24, 26, 34, 36, 38$ it contains $2, 2, 6, 6, 6, 2, 2$ vectors. Each violates (10), in every case but one already with $p = 1$; the exception is $n = 26$, where two of the six vectors need $p = 2$. The certificate `anc/kstar_certificates.json` lists the reduced bases and, for each vector, a violated instance (p, J).

Lower bounds for $14 \le n \le 40$. For each n we found a witness for $k^*(n) \ge K$, with K the value of Table 2: an arrangement with all row sums $M(n)$ whose column deviations satisfy $(t-1)^K \mid P_\gamma$ and $\gamma \neq 0$; each was verified in exact integer arithmetic. □

n	14	15	16	17	18	19	20	21	22	23	24	25	26	27
$k^*(n)$	12	13	14	15	15	16	16	17	17	18	18	19	19	20
$\max_j \lvert\gamma_j\rvert$	297	572	1001	2002	1430	2860	1794	3094	2210	3638	2380	4522	2736	4541
$n^2(n-1)/2$	1274	1575	1920	2312	2754	3249	3800	4410	5082	5819	6624	7500	8450	9477

n	28	29	30	31	32	33	34	35	36	37	38	39	40
$k^*(n)$	21	21	22	22	23	23	23	24	24	25	25	26	26
$\max_j \lvert\gamma_j\rvert$	8892	6405	11737	9330	14881	11799	9177	18147	14766	24150	16515	26520	22233
$n^2(n-1)/2$	10584	11774	13050	14415	15872	17424	19074	20825	22680	24642	26714	28899	31200

Table 2: Exact values of the permutation threshold for $14 \le n \le 40$ (Theorem 3.7(c)), the largest column deviation $\max_j |\gamma_j|$ of the witness arrangement, and the bound $n^2(n-1)/2$ of Lemma 3.8. The upper bounds for $18 \le n \le 40$ were computed by two independent exact lattice enumerations (`anc/kstar_verify.py`, `anc/kstar_enum_independent.py`) and, for $n \le 33$, also by integer programming (`anc/kstar_crosscheck/`); a certificate is in `anc/kstar_certificates.json`.

The witnesses. The target vectors γ were chosen among the shortest vectors of $\Lambda_{n,k^*(n)-1}$ satisfying (10), as produced by the enumeration. Starting from a semi-magic square of Lemma 3.12 (or a Siamese (de la Loubère) magic square [3]), a steepest-descent search over exchanges within a row, which fix the row sums, minimised $\sum_j (\text{column sum}_j - M(n) - \gamma_j)^2$, with random exchanges to leave local minima; for the hardest cases a simulated annealing over arbitrary exchanges, with energy the squared distance of both row and column sums to their targets, was used. For each $14 \le n \le 40$ the witness search succeeded at the last k at which $\Lambda_{n,k}$ still contains a nonzero vector satisfying (10), so $k^*(n)$ equals the least k for which $\Lambda_{n,k}$ contains no such vector. We do not know whether this holds for all n. Some witnesses come close to the bound of Lemma 3.8: $\max_j |\gamma_j| = 924$ against 1014 at $n = 13$, and 24150 against 24642 at $n = 37$. Beyond $n = 40$

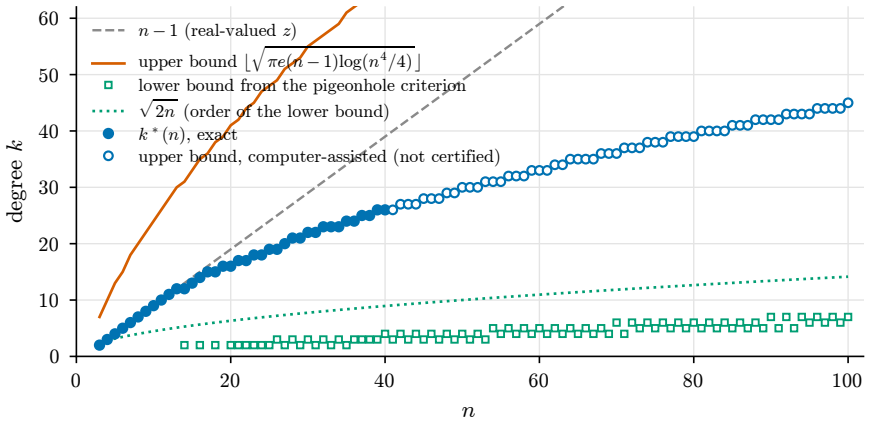


Figure 2: The permutation threshold. Exact values of $k^*(n)$ for $3 \le n \le 40$ (dots), computer-assisted upper bounds for $41 \le n \le 100$ (two independent implementations agree, not certified; open circles), the linear-algebra threshold $n-1$, the bound of Theorem 3.7(a), the lower bound given by (11) (open squares, shown where it gives $K \ge 2$; for odd n with small r_q it is trivial and is not plotted), and $\sqrt{2n}$, the asymptotic order of (b), which (11) approaches slowly.

we computed only upper bounds (computer-assisted; two independent implementations agree for every $n \le 100$, but the values carry no certificate; Figure 2); for instance $k^*(50) \le 30$, $k^*(80) \le 39$ and $k^*(100) \le 45$.

4 The energy of a random arrangement

Let S be a uniformly random arrangement: $z_c = b_{\pi(c)}$ with π uniform and b the centred values $b_v = v - \frac{N+1}{2}$, $1 \le v \le N$. For a test function φ put $T_\varphi = \langle \varphi, z\rangle = \sum_c a_c z_c$ with $a_c = \tilde\varphi_c/N$, a linear permutation statistic (Section 2.3); recall that $\tilde\varphi = \varphi - \bar\varphi\mathbf{1}$ and $\sigma_n^2 = (n^2+1)/12$ (7). Throughout this section $\mathbb{E}$, Var and Cov refer to the random arrangement.

4.1 Means

Proposition 4.1 (Means and covariance structure)**.** *For all test functions φ, ψ, $\mathbb{E}[T_\varphi] = 0$ and $\mathbb{E}[T_\varphi T_\psi] = \sigma_n^2\langle\tilde\varphi, \tilde\psi\rangle$. Consequently*

$$\mathbb{E}\,\langle x, z\rangle^2 = \mathbb{E}\,\langle y, z\rangle^2 = \frac{n^4-1}{144}, \qquad \mathbb{E}\,\langle \mathbf{1}_L, z\rangle^2 = \frac{(n-1)(n^2+1)}{12\,n^2} \quad \text{for every line } L, \tag{12}$$

$$\mathbb{E}[E_4] = \frac{(n-1)(n^2+1)(n^3+n^2+12)}{72\,n^2}, \qquad \mathbb{E}[E_{\text{full}}] = \frac{n^4-1}{6\,n^2}. \tag{13}$$

The covariance matrix of the vector of statistics in E_4 (resp. E_{full}) is $\sigma_n^2\Gamma$, with $\Gamma = (\langle\tilde\varphi_\ell, \tilde\varphi_{\ell'}\rangle)_{\ell,\ell'}$ the Gram matrix of the centred test functions φ_ℓ of the energy; its eigenvalues are

$$E_4:\ \sigma_n^2\Big\{\tfrac{n^2-1}{12}\,(\times 2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\}, \qquad E_{\text{full}}:\ \sigma_n^2\Big\{\tfrac{1}{n}\,(\times 2n-2),\ \tfrac{n-1}{n^2}\,(\times 2),\ 0\,(\times 2)\Big\} \tag{14}$$

for odd n; for even n the pair $\{\frac{n-1}{n^2}, \frac{n-1}{n^2}\}$ is replaced by $\{\frac{1}{n}, \frac{n-2}{n^2}\}$.

Proof. $\sum_c a_c = 0$ gives $\mathbb{E}T_\varphi = 0$, and (6) with $\sum_v b_v^2 = N\|z\|^2$ gives $\mathbb{E}[T_\varphi T_\psi] = \frac{1}{N-1}\cdot\frac{\langle\tilde\varphi,\tilde\psi\rangle}{N}\cdot N\|z\|^2 = \sigma_n^2\langle\tilde\varphi,\tilde\psi\rangle$. The entries of Γ follow from Lemma 2.3: $\|\tilde{\mathbf{1}}_L\|^2 = (n-1)/n^2$, two lines meeting in r cells have $\langle\tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'}\rangle = r/N - 1/n^2$, and x, y are orthogonal to each other and to the centred diagonal indicators, since a diagonal meets every row and every column once. The rows give the block $\frac{1}{n}(I - \frac{1}{n}J)$, the columns likewise, and the means are the traces. □

The means are elementary (sampling without replacement): a line sum is the sum of a sample of size n drawn without replacement from $\{1,\dots,N\}$, so $\mathbb{E}\delta_L^2 = n\|z\|^2\frac{N-n}{N-1}$. The two means differ

only through the x, y terms of E_4 and the $2n$ row and column terms of E_{full}, whose means are $(n^4-1)/72$ and $(n-1)(n^2+1)/(6n)$. These agree if and only if $n(n+1)=12$, i.e. $n=3$, where both energies have mean $40/27$.

Corollary 4.2 (Mean and variance of Loly's invariant)**.** *For a uniformly random arrangement, $I(S)$ of Corollary 3.5 satisfies*

$$\mathbb{E}[I(S)] = \frac{n^2(n^4-1)}{12}, \qquad \operatorname{Var}\big(I(S)\big) = N^2\sigma_n^2 \left\|\widetilde{x^2+y^2}\right\|^2 = \frac{n^4(n^4-1)(n^2-4)}{1080}. \tag{15}$$

The mean equals Loly's value $n^2(n^4-1)/12$ for every random arrangement with $\mathbb{E}z_c = 0$ for every cell (for instance any distribution invariant under complementation $s \mapsto N+1-s$); magic and semi-magic squares take this value exactly.

Proof. By (9), $I(S) - \frac{n^2(n^4-1)}{12} = N\,T_{x^2+y^2}$, so Proposition 4.1 gives the mean and $\operatorname{Var}(I) = N^2\sigma_n^2\|\widetilde{x^2+y^2}\|^2$. Since x^2 depends only on the column and y^2 only on the row, $\widetilde{x^2} \perp \widetilde{y^2}$, so $\|\widetilde{x^2+y^2}\|^2 = 2\big(\langle x^4,\mathbf{1}\rangle - \langle x^2,\mathbf{1}\rangle^2\big)$ with $\langle x^2,\mathbf{1}\rangle = \frac{n^2-1}{12}$ and $\langle x^4,\mathbf{1}\rangle = \frac{(n^2-1)(3n^2-7)}{240}$, which gives $\frac{(n^2-1)(n^2-4)}{90}$. □

For $n=3$, $\operatorname{Var}(I)=30$ and I takes 29 values from 46 to 74; for $n=4$, $\operatorname{Var}(I)=2176/3$ and $\Pr(I=340) = 13049/450450 \approx 0.029$ (`anc/verify_numbers.py`).

4.2 Variances

Write a random energy as a quadratic form $E = \sum_\ell T_{\varphi_\ell}^2 = z^{\mathsf{T}}Az$ with $A = \sum_\ell a_\ell a_\ell^{\mathsf{T}}$, where $a_\ell = \tilde{\varphi}_\ell/N$. A is symmetric with zero row sums.

Proposition 4.3 (Mantel [22], Hubert [16]; specialised to zero row sums and centred values)**.** *Let A be a symmetric $N\times N$ matrix with zero row sums, $N\ge 4$, and let z be a uniformly random permutation of the centred values $b_v = v - \frac{N+1}{2}$, $1\le v\le N$. Then $\mathbb{E}[z^{\mathsf{T}}Az] = \frac{N(N+1)}{12}\operatorname{tr}A$ and*

$$\operatorname{Var}(z^{\mathsf{T}}Az) = -\frac{N(N+1)(2N+3)}{360}(\operatorname{tr}A)^2 + \frac{N(N^2-1)(5N+6)}{360}\operatorname{tr}(A^2) - \frac{N^2(N+1)^2}{120}\sum_a A_{aa}^2. \tag{16}$$

Proof (sketch; details in Appendix C). Expand $\mathbb{E}[(z^{\mathsf{T}}Az)^2] = \sum_{a,b,c,d} A_{ab}A_{cd}\mathbb{E}[z_az_bz_cz_d]$ according to the equality pattern of (a,b,c,d), a set partition of $\{1,2,3,4\}$, and apply Möbius inversion on the partition lattice, so that each pattern contributes an unrestricted sum. An unrestricted sum in which some index occurs in only one factor $A_{\cdot\cdot}$ is a row sum of A, hence 0. Only the four partitions without singletons survive, 12|34, 13|24, 14|23 and 1234, contributing $(\operatorname{tr}A)^2$, $\operatorname{tr}A^2$, $\operatorname{tr}A^2$ and $\sum_a A_{aa}^2$, with coefficients given by the joint moments of a random permutation of the centred values. □

For the energies the three invariants are explicit; this is where the parity of n enters, through the centre cell shared by the two diagonals when n is odd.

Theorem 4.4 (Variances of the energies)**.** *For every $n\ge 3$,*

$$\operatorname{Var}(E_4) = \begin{cases} \dfrac{(n-1)(n^2+1)\,V_{\mathrm{o}}(n)}{129600\,n^6} & n \text{ odd}, \\[2ex] \dfrac{(n^2+1)\,V_{\mathrm{e}}(n)}{129600\,n^4} & n \text{ even}, \end{cases} \qquad \operatorname{Var}(E_{\text{full}}) = \begin{cases} \dfrac{(n-1)(n^2+1)\,G_{\mathrm{o}}(n)}{180\,n^6} & n \text{ odd}, \\[2ex] \dfrac{(n-2)(n^2+1)\,G_{\mathrm{e}}(n)}{180\,n^5} & n \text{ even}, \end{cases}$$

where

$$V_{\mathrm{o}} = 25n^{11} + 25n^{10} - 82n^9 - 82n^8 - 9n^7 - 1209n^6 + 3738n^5 - 5862n^4 + 2160n^3 - 2880n^2 - 4320n + 4320,$$
$$V_{\mathrm{e}} = 25n^{10} - 107n^8 + 73n^6 - 1200n^5 + 4947n^4 - 9600n^3 + 13782n^2 - 13680n + 10080,$$
$$G_{\mathrm{o}} = 5n^6 - 5n^5 - 17n^4 + 11n^3 - 24n^2 + 18n + 6, \qquad G_{\mathrm{e}} = 5n^5 - 12n^3 + 12n^2 - 23n + 12.$$

For both parities,

$$\mathrm{Var}(E_4) = \frac{n^8}{5184} - \frac{41\,n^6}{64800} + O(n^4), \qquad \mathrm{Var}(E_{\mathrm{full}}) = \frac{n^3}{36} - \frac{n^2}{18} - \frac{7n}{180} + O(1).$$

Proof. Proposition 4.3 with $N = n^2$ and the invariants (18)–(19) of Appendix C; the expansions follow. □

At $n = 3$ the formulas give $\mathrm{Var}(E_4) = 1532/2187$ and $\mathrm{Var}(E_{\mathrm{full}}) = 398/2187$, the values obtained by enumerating all 9! arrangements.

4.3 Limit law

Corollary 4.5 (Limit law)**.** *Let $\varphi_1 = x$, $\varphi_2 = y$, $\varphi_3 = \mathbf{1}_{D_{\mathrm{m}}}$, $\varphi_4 = \mathbf{1}_{D_{\mathrm{a}}}$, $T_\ell = T_{\varphi_\ell}$, and let $\tau_\ell^2 = \mathrm{Var}T_\ell$ (the diagonal of $\sigma_n^2\Gamma$). As $n \to \infty$, $\big(T_1/\tau_1, \dots, T_4/\tau_4\big)$ converges in distribution to a standard Gaussian vector in $\mathbb{R}^4$, and*

$$\frac{E_4}{\mathbb{E}[E_4]} \xrightarrow{\;d\;} \mathrm{Exp}(1). \tag{17}$$

Proof. Fix $\alpha \in \mathbb{R}^4 \setminus \{0\}$ and let $a = \sum_\ell \alpha_\ell a_\ell/\tau_\ell$ be the corresponding combination of the four vectors $a_\ell = \tilde{\varphi}_\ell/N$. Then $\sum_\ell \alpha_\ell T_\ell/\tau_\ell = \sum_c a_c z_c = \sum_c a_c b_{\pi(c)}$ with $\sum_c a_c = \sum_v b_v = 0$, and

$$\left(N\,\frac{\max_c a_c^2\,\max_v b_v^2}{\sum_c a_c^2\,\sum_v b_v^2}\right)^{1/2} = \frac{\max_c |a_c|}{\|a\|_2} \cdot \frac{\max_v |b_v|}{(\sum_v b_v^2/N)^{1/2}}$$

is the square root of the quotient in (8). The second factor tends to $\sqrt{3}$. For the first, $\max_c |a_c| \le \sum_\ell |\alpha_\ell| \max_c |a_{\ell,c}|/\tau_\ell$, and $\max_c |a_{\ell,c}|/\|a_\ell\|_2$ is $O(n^{-1})$ for x and y and $O(n^{-1/2})$ for a diagonal indicator (a centred indicator of n of the N cells). Since $\mathrm{Var}T_\ell = \sigma_n^2 N \|a_\ell\|_2^2$ by (6), $\|a_\ell\|_2/\tau_\ell = (\sigma_n\sqrt{N})^{-1}$ for every ℓ, so $\|a\|_2^2 = \alpha^{\mathsf{T}} C_n \alpha/(N\sigma_n^2)$ with C_n the correlation matrix of $(T_1, \dots, T_4)$, which tends to the identity because the only nonzero off-diagonal correlation is $\mathrm{corr}(T_3, T_4) = -1/(n-1)$ for even n. Hence $\max_c |a_c|/\|a\|_2 \le \sum_\ell |\alpha_\ell| \big(\max_c |a_{\ell,c}|/\|a_\ell\|_2\big)/(\alpha^{\mathsf{T}} C_n \alpha)^{1/2} \to 0$, condition (8) holds, [15, Thm. 4] with its condition (14) gives asymptotic normality of every such combination, and the Cramér–Wold device gives the joint limit $\mathcal{N}(0, I_4)$. In $E_4 = \tau_1^2\big((T_1/\tau_1)^2 + (T_2/\tau_2)^2\big) + \tau_3^2\big((T_3/\tau_3)^2 + (T_4/\tau_4)^2\big)$ (note $\tau_1 = \tau_2$ and $\tau_3 = \tau_4$) the diagonal weights satisfy $\tau_3^2/\tau_1^2 = 12/(n^2(n+1)) \to 0$, so $E_4/(2\tau_1^2) \to \frac{1}{2}\chi_2^2 = \mathrm{Exp}(1)$; and $\mathbb{E}[E_4]/(2\tau_1^2) \to 1$. □

Remark 4.6 (The χ^2-mixture approximation at finite n). The χ^2-mixture approximation, which treats the statistics of an energy as jointly Gaussian, replaces the energy by $\sum_j w_j \chi_{1,j}^2$, with independent χ_1^2 variables and weights w_j the eigenvalues (14); its density can be computed by inversion of the characteristic function [17] (Figure 3). It has the exact mean but variance $2\sum_j w_j^2$, and by Theorem 4.4 the ratio of the exact variance to this value is

$$1 - \frac{2}{n} - \frac{7}{5n^2} + O(n^{-3}) \quad (E_{\mathrm{full}}), \qquad 1 - \frac{82}{25n^2} + O(n^{-3}) \quad (E_4).$$

The χ^2-mixture variance is $2\,\mathrm{tr}\big((\sigma_n^2\Gamma)^2\big) = N^2(N+1)^2\,\mathrm{tr}(A^2)/72$. In (16) the coefficient of $\mathrm{tr}(A^2)$ is this value times $1 - (4N+6)/(5N(N+1)) = 1 - 4/(5n^2) + O(n^{-4})$, and the $(\mathrm{tr}\,A)^2$ and $\sum_a A_{aa}^2$ terms are negative. For E_{full} they give the leading deficit $-4/(5n) - 6/(5n) = -2/n$; for E_4 all three terms contribute only at order n^{-2} ($-4/5$, $-4/5$, $-42/25$, total $-82/25$). For E_{full} the number of statistics grows with n and Corollary 4.5 does not apply. The approximation is poor for small n: at $n = 3$ the χ^2-mixture standard deviation of E_{full} is about twice the exact one (Table 3).

n	energy	$\mathbb{E}[E]$	$\sqrt{\mathrm{Var}E}$	χ^2-mixture sd	variance ratio
3	E_4	40/27	0.8370	1.171	0.511
3	E_{full}	40/27	0.4266	0.869	0.241
4	E_4	391/96	3.1284	3.586	0.761
4	E_{full}	85/32	0.9074	1.349	0.453
5	E_4	234/25	8.0483	8.694	0.857
5	E_{full}	104/25	1.3966	1.867	0.560
6	E_4	2035/108	17.122	18.007	0.904
6	E_{full}	1295/216	1.9671	2.459	0.640
7	E_4	5050/147	32.174	33.349	0.931
7	E_{full}	400/49	2.5697	3.089	0.692
8	E_4	22295/384	55.377	56.888	0.948
8	E_{full}	1365/128	3.2352	3.777	0.734

Table 3: Exact means and standard deviations of the two energies (Proposition 4.1, Theorem 4.4), the standard deviation of the χ^2-mixture approximation of Remark 4.6, and the ratio of the exact to the χ^2-mixture variance.

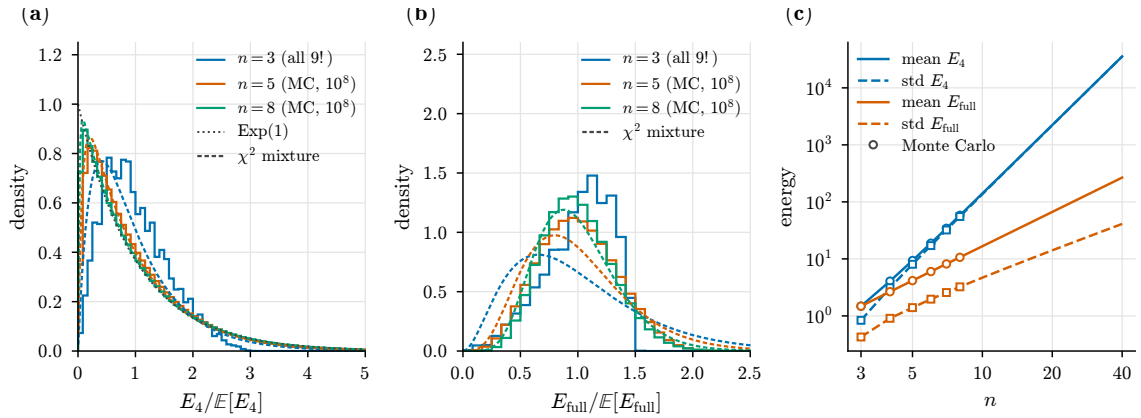


Figure 3: Energy of a uniformly random arrangement. (a) Distribution of $E_4/\mathbb{E}[E_4]$ for $n = 3$ (all 9! arrangements) and $n = 5, 8$ (10^8 Monte Carlo samples each), with the densities of the χ^2-mixture approximation (dashed) and the limit Exp(1) of Corollary 4.5 (dotted). (b) The same for E_{full}, where the χ^2-mixture approximation improves slowly (Remark 4.6). (c) Exact mean (solid) and standard deviation (dashed) of both energies for $3 \le n \le 40$, with Monte Carlo values for $3 \le n \le 8$ (open markers).

5 Computations for small orders

All counts below were computed in exact integer arithmetic; the methods are described in Appendix A, and every number is produced by a script in the ancillary directory `anc/`.

5.1 The classes L_3 and L_4

Let L_n be the class of arrangements whose first moments vanish and whose diagonal sums equal $M(n)$, i.e. the arrangements with $E_4 = 0$: $\langle x, z\rangle = \langle y, z\rangle = 0$ and both diagonal sums equal to $M(n)$. With cells indexed by (i, j), these are the conditions $U_1 = U_2 = \frac{n-1}{2} \cdot \frac{N(N+1)}{2}$ and $U_3 = U_4 = M(n)$ on the sums $U_1 = \sum_{i,j} j\, s_{ij}$, $U_2 = \sum_{i,j} i\, s_{ij}$, $U_3 = \sum_i s_{ii}$, $U_4 = \sum_i s_{i,n-1-i}$, which are additive over cells.

Proposition 5.1. *For $n = 3$, $E_4(S) = 0$ if and only if S is magic; that is, L_3 consists of the eight magic squares.*

Proof. Write the centred entries $z_c = s_c - 5 \in \{-4, \dots, 4\}$ as $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, so that $\sum_c z_c = 0$, $\sum_c z_c^2 = 60$, and every line sum of z is the deviation of that line. The degree-1 conditions with $\sum_j \gamma_j = \sum_i \rho_i = 0$ give $\gamma = (\xi, -2\xi, \xi)$ and $\rho = (\eta, -2\eta, \eta)$, and the diagonal conditions give $a + e + i =$

$c+e+g=0$. The middle row, the middle column and the two diagonals cover every cell once and the centre three more times, so $3e=-2(\xi+\eta)$; as e is an integer, $\xi+\eta=3\kappa$ and $e=-2\kappa$ with $\kappa\in\mathbb{Z}$. Put $a'=a-\kappa$, $c'=c-\kappa$, $\xi'=\xi-\kappa$, $\eta'=\eta-\kappa$, so that $\xi'+\eta'=\kappa$. The diagonals give $i=\kappa-a'$, $g=\kappa-c'$; the first and last rows (sum η) and columns (sum ξ) then give

$$z=\begin{pmatrix}\kappa+a' & -\xi'-a'-c' & \kappa+c'\\ -\eta'-a'+c' & -2\kappa & -\eta'+a'-c'\\ \kappa-c' & -\xi'+a'+c' & \kappa-a'\end{pmatrix},\qquad \sum_c z_c^2=2\big(4\kappa^2+3(a'^2+c'^2)+\xi'^2+\eta'^2\big),$$

so $4\kappa^2+3(a'^2+c'^2)+\xi'^2+\eta'^2=30$. The four corners are distinct, so $a'\neq 0$, $c'\neq 0$ and $|a'|\neq|c'|$, hence $a'^2+c'^2\geq 5$; therefore $4\kappa^2+\xi'^2+\eta'^2$ is a multiple of 3 and at most 15, and $a'^2+c'^2=10-(4\kappa^2+\xi'^2+\eta'^2)/3$. So $|\kappa|\leq 1$. If $|\kappa|=1$, then $\xi'+\eta'=\pm1$ makes $\xi'^2+\eta'^2$ odd, so $4\kappa^2+\xi'^2+\eta'^2\in\{9,15\}$ (as it is at least 5) and $\xi'^2+\eta'^2\in\{5,11\}$; 11 is not a sum of two squares, and $\xi'^2+\eta'^2=5$ gives $a'^2+c'^2=7$, which is not either. So $\kappa=0$, $\eta'=-\xi'$, and $2\xi'^2\leq 15$ divisible by 3 forces $\xi'=0$. Hence $\xi=\eta=0$: all rows, columns and diagonals have deviation 0, and S is magic. The converse is clear. □

Remark 5.2. The enumeration of all 9! arrangements confirms Proposition 5.1 and each step of its proof (`anc/n3_lowmode_proof.py`): exactly 760 arrangements satisfy the degree-1 conditions, 4,224 have both diagonal sums 15, 72 (the semi-magic squares) satisfy the degree-2 conditions, and 8, the magic squares, satisfy the degree-1 and both diagonal conditions; the only integer solution of $4\kappa^2+3(a'^2+c'^2)+\xi'^2+\eta'^2=30$ with $\xi'+\eta'=\kappa$, $a'c'\neq 0$, $|a'|\neq|c'|$ is $\kappa=\xi'=\eta'=0$, $a'^2+c'^2=10$. The four conditions are weaker than the magic conditions for real-valued z; the proof uses that the entries are distinct integers with $\sum_c z_c^2=60$.

For $n=4$ the equivalence fails (Remark 2.5).

Proposition 5.3 (The size of L_4)**.** $|L_3|=8$ *(Proposition 5.1), and*

$$|L_4|=14{,}639{,}440=8\times 1{,}829{,}930,$$

of which 7,040 *are magic squares. The class* $\{\langle x,z\rangle=\langle y,z\rangle=0\}$ *has* 6,697,177,424 *members for* $n=4$*, and the class with both diagonal sums equal to* 34 *has* 49,932,288,000*.*

Computation. A meet-in-the-middle enumeration over $(U_1,\dots,U_4)$, splitting the grid into the top two and bottom two rows (Appendix A); the matching pairs were expanded to obtain the breakdown of Table 4. The same program reproduces the brute-force counts for $n=3$ (8, 760, 4224, and 72 semi-magic squares) and, with the $2n$ line sums in place of U_1,U_2, the 7,040 magic and 549,504 semi-magic squares of order 4. Two further programs written independently of `anc/lowmode_enum.c` (`anc/lowmode_recount.py`, `anc/table4_recount.py`) together reproduce $|L_4|$, the first-moment count 6,697,177,424 and every entry of Table 4. The diagonal-only count also follows from a direct count: there are 2150 ordered pairs of disjoint 4-subsets of $\{1,\dots,16\}$ with sum 34, and $2150\cdot(4!)^2\cdot 8!=49{,}932{,}288{,}000$. □

Since D_4 acts freely on the set of all arrangements (entries are distinct) and preserves L_4, L_4 consists of 1,829,930 orbits of size 8.

Exactly two or three rows with sum 34 are impossible in L_4: the row deviations satisfy $\sum_i\rho_i=0$ and $\sum_i i\rho_i=0$, which cannot hold with exactly one or two nonzero ρ_i; the same holds for columns. Thus $7040/14{,}639{,}440\approx 0.048\%$ of L_4 is magic, and 9,637,312 members (66%) have no row or column with sum 34, among them the arrangement of Remark 2.5.

A local-limit heuristic. Under a uniformly random arrangement the vector $(U_1,\dots,U_4)$ has mean equal to the target vector and covariance matrix Σ_U given by (6): U_1,U_2 are uncorrelated with each other and with the diagonal sums, which have correlation $-1/(n-1)$ for even n and 0 for odd n. Since any two arrangements differ by a product of transpositions, the differences

	columns with sum 34			
rows with sum 34	0	1	4	total
0	9 637 312	2 172 824	100 728	11 910 864
1	2 172 824	429 312	9 336	2 611 472
4	100 728	9 336	**7 040** (magic)	117 104
total	11 910 864	2 611 472	117 104	14 639 440

Table 4: The members of L_4 by the number of rows and of columns with sum $M(4) = 34$; the table is symmetric under transposition.

of $(U_1, \dots, U_4)$ generate the same lattice as the increments of single transpositions; by a Smith normal form computation ($n \le 8$) this lattice is $\mathbb{Z}^4$. A Gaussian local limit would therefore give the heuristic estimate $|L_n| \approx N!\,(2\pi)^{-2}\det(\Sigma_U)^{-1/2}$. This estimate is 13.6 for $n = 3$ ($|L_3| = 8$) and 1.82×10^7 for $n = 4$, so $|L_4|$ is 0.80 times the estimate; for the two-condition classes the ratios at $n = 3, 4$ are $0.59, 0.91$ (first moments) and $1.10, 0.96$ (diagonals). For $n = 5, 6$ Monte Carlo (10^{10} and 2×10^{10} samples, 328 and 62 members of L_n) gives $|L_5| \approx 5.1\times10^{17}$ ($\pm6\%$) and $|L_6| \approx 1.2\times10^{33}$ ($\pm13\%$), ratios 0.76 and 0.78 to the estimate. Since $\det(\Sigma_U)^{1/2}$ grows like n^{13}, the heuristic suggests $|L_n|/N! \asymp n^{-13}$. The same heuristic applied to the $2n$ independent line conditions, each with variance $\asymp n^5/12$, suggests that magic squares form a fraction $\exp(-5n\log n + O(n))$ of all arrangements, hence a vanishing proportion of L_n. Neither statement is proved here; local limit theorems for permutation statistics of this kind are studied in [18, 11], and related counts of matrices with prescribed margins in [10].

5.2 Single swaps

Remark 5.4 (Single-swap formula). Let S^{ab} be obtained from S by exchanging the entries of cells $a \ne b$, let $d = s_b - s_a$, let $\lambda(a, b)$ be the number of lines containing exactly one of a, b, and $u_S(c) = \sum_{L \ni c} \delta_L(S)$. Expanding (5),

$$N^2\big(E_{\text{full}}(S^{ab}) - E_{\text{full}}(S)\big) = d^2\lambda(a,b) + 2d\big(u_S(a) - u_S(b)\big),$$

the incremental evaluation used in local search [29]. Here $\lambda(a, b) = 2[i \ne i'] + 2[j \ne j'] +$ [exactly one of a, b lies on D_{m}] + [exactly one of a, b lies on D_{a}] $\in \{2, \dots, 6\}$ for $a = (i, j)$, $b = (i', j')$. For a magic square S, $N^2E_{\text{full}}(S^{ab}) = d^2\lambda(a, b) \ge 2$, so no two magic squares differ by a transposition, and the increase of N^2E_{full} under a transposition lies in the set $\{d^2\lambda : 1 \le d \le N-1,\ 2 \le \lambda \le 6\}$. The smallest increase $g_{\min}(S) = \min_{a\ne b} d^2\lambda(a, b)$ satisfies $2 \le g_{\min}(S) \le 6$: the value 2 occurs if and only if two consecutive values lie in a common row or column and off both diagonals, and consecutive values give at most 6.

Call S a local minimum with respect to transpositions if no transposition lowers the energy. For $n = 3$ the local minima of E_{full} are the eight magic squares and eight non-magic arrangements with $N^2E_{\text{full}} = 2$, all strict; E_4 has 344, of which 336 are non-magic, and only 72 of them (64 non-magic) are strict. Steepest descent on E_{full} (at each step perform the transposition with the most negative change of N^2E_{full}, ties broken by the first cell pair in row-major order; stop when no transposition lowers it) reaches a magic square in 19.5% ($\pm0.1\%$), 1.2% and 0.8% of 2×10^5 runs from uniformly random starts for $n = 4, 5, 6$ (`anc/algorithms.py`).

6 Open questions

The main results are the two-sided bound $(\sqrt{2} - o(1))\sqrt{n} \le k^*(n) \le (2\sqrt{\pi e} + o(1))\sqrt{n\log n}$ for the permutation threshold, with exact values for $n \le 40$ (computer-assisted for the upper bounds when $18 \le n \le 40$), and exact second moments of the line-sum energies of a random arrangement. The following questions remain.

1. *The order of $k^*(n)$.* Is $k^*(n) \asymp \sqrt{n}$? The upper bound uses only that γ is a nonzero integer vector with $\|\gamma\|_1 \le n^4/4$. In that class the pigeonhole principle produces, for all large n, vectors vanishing to order $\ge \lfloor 2\sqrt{n} \rfloor$ at 1 (Section 3.3), so a bound of smaller order than $\sqrt{n}$ needs more information about γ. Whether the order $\sqrt{n \log n}$ of Lemma 3.9 is attained within this class is, as far as we know, open; this is separate from realisability, i.e. which such vectors are column deviations of arrangements. For $14 \le n \le 40$, $k^*(n)$ is the least k for which $\Lambda_{n,k}$ contains no nonzero vector satisfying (10) (Section 3.5). Is this true for all n, and is $k^*(n)$ nondecreasing?

2. *The size of L_n.* Is $|L_n| \sim N!\,(2\pi)^{-2} \det(\Sigma_U)^{-1/2}\, c_n$ with c_n bounded away from 0 and ∞, i.e. does a local limit theorem hold for the four statistics $(U_1, \dots, U_4)$ of Section 5.1? The ratio c_n is 0.59, 0.80 for $n = 3, 4$ and about 0.76, 0.78 for $n = 5, 6$.

3. *Other families.* The orthogonality formulation applies to pandiagonal and Franklin squares [20], with W replaced by the span of the relevant line indicators. The corresponding thresholds and random energies have not been worked out.

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A Computational methods

All computations were run on an Apple M4 Max under macOS with Python 3.9 (NumPy, SciPy, SymPy) and C compiled with Apple clang; uniformly random arrangements were generated by Fisher–Yates shuffles. The scripts and data are in `anc/`, with a `README` listing the commands that regenerate every table and figure.

The permutation threshold. `anc/kstar_verify.py` checks in exact integer arithmetic that each witness arrangement ($5 \le n \le 40$) is a permutation of $1, \dots, N$ with all row sums $M(n)$, and computes the order of vanishing of P_γ at 1 from the binomial moments $\sum_j \binom{j}{r}\gamma_j$. For the upper bounds it reduces the basis $\{t^i(t-1)^{k+1}\}$ of $\Lambda_{n,k}$ by LLL (exact integer arithmetic) and enumerates all lattice vectors with $\|\gamma\|_2^2 \le n^5(n^2-1)/12$ by the Fincke–Pohst method with exact rational Gram–Schmidt coefficients, testing (10) for every p. Since $\Lambda_{n,k+1} \subset \Lambda_{n,k}$, the existence of a nonzero vector satisfying (10) is monotone in k, and the least k for which $\Lambda_{n,k}$ contains no such vector is the upper bound. A second implementation, written independently (`anc/kstar_enum_independent.py`, with its own exact LLL reduction and enumeration), reproduces the upper bounds for $14 \le n \le 40$ and writes the certificate `anc/kstar_certificates.json`: for each n the reduced basis of $\Lambda_{n,k^*(n)}$, the exact squared Gram–Schmidt norms, and every nonzero vector of the ball with a violated instance (p, J) of (10). The short script `anc/kstar_check_certificates.py` checks in exact arithmetic that each basis lies in $\Lambda_{n,k^*(n)}$ and has the same Gram determinant as $\{t^i(t-1)^{k^*(n)+1}\}$, hence generates it; recomputes the Gram–Schmidt norms (when all of them exceed the radius, the ball contains no nonzero lattice vector); and checks each violated instance. For $n \le 33$ the upper bounds were also reproduced by floating-point integer programming (`anc/kstar_crosscheck/`). The witnesses were found with `anc/realise.py` and `anc/anneal2.c`, starting from the squares of Lemma 3.12 (`anc/constructions.py`; an independent implementation of the construction is `anc/twin_squares.py`).

Variances. `anc/variance_proof.py` derives (16) symbolically by Möbius inversion over the 15 set partitions of $\{1, 2, 3, 4\}$, checks it by enumerating all permutations for random matrices with $N \le 7$, computes the invariants (18)–(19) symbolically and checks them against the explicit matrices for $3 \le n \le 25$, and compares the result with the enumeration of all 9! arrangements for $n = 3$. Monte Carlo with 10^8 arrangements per order ($3 \le n \le 8$) agrees with the exact means and standard deviations to within 5×10^{-4} relative error (`anc/mc_results.json`). The densities in Figure 3 were computed by numerical inversion of the characteristic function [17].

Meet-in-the-middle enumeration of L_4. The 16 cells are split into the top two rows A and the bottom two rows B. For each of the $\binom{16}{8} = 12{,}870$ value sets V and each of the 8! bijections $V \to A$ the partial vector $(U_1, \dots, U_4)|_A$ is packed into an integer key; for the complementary set and each bijection $V^c \to B$ the key of the target minus the partial vector is formed; the two key lists are sorted and merged, and the sum over matching keys of the products of multiplicities is $|L_4|$. This touches about 1.04×10^9 partial vectors instead of $16! \approx 2.1 \times 10^{13}$ arrangements (`anc/lowmode_enum.c`). The NumPy programs `anc/lowmode_recount.py` and `anc/table4_recount.py` use the same split with a different packing of the keys and tag each half by its number of rows (and the sum of a chosen column) equal to 34; together with a direct enumeration of the members of L_4 whose four rows all sum to 34, they recount $|L_4|$, the first-moment class and every entry of Table 4.

`anc/verify_numbers.py` and `anc/balance_check.py` recompute the remaining numbers (the former in exact arithmetic).

B Squares with many complementary rows

Proof of Lemma 3.12(1). Give the value v the weight $|2v-(N+1)|$, an odd number in $\{1, 3, \dots, N-1\}$; each weight w belongs to exactly the two complementary values $(N+1 \pm w)/2$. Index the weights by $\iota = (w-1)/2 \in \{0, \dots, 2h^2-1\}$, $w_\iota = 2\iota + 1$. It suffices to partition the indices into h sets $R_0, \dots, R_{h-1}$ of size n and choose signs $\sigma_\iota \in \{\pm 1\}$ with $\sum_{\iota \in R_r} \sigma_\iota w_\iota = 0$ for every r: then row r receives the values $(N+1+\sigma_\iota w_\iota)/2$, $\iota \in R_r$, in any order, and row $h+r$ their complements in the same columns. Every value occurs once, row r sums to $n\frac{N+1}{2} = M(n)$, row $h+r$ to $n(N+1) - M(n) = M(n)$, and every column to $h(N+1) = M(n)$. A *domino* $\{\iota, \iota+1\}$ (for odd h, any two consecutive unused indices; they need not be of the form $\{2t, 2t+1\}$) with signs $(+,-)$ or $(-,+)$ contributes ∓ 2. If h is even, give each row h of the dominoes $\{2t, 2t+1\}$ with alternating sign patterns. If $h \geq 7$ is odd, give row r the four indices $4r, \dots, 4r+3$ with signs $(+,+,+,-)$, contributing $(8r+1) + (8r+3) + (8r+5) - (8r+7) = 16r+2$; the two indices $\alpha_r = 5h-1-r$ and $\beta_r = 5h+7r$ with signs $(+,-)$, contributing $2(\alpha_r - \beta_r) = -16r-2$; and $h-3$ dominoes with alternating sign patterns. The quadruples fill $[0, 4h-1]$ and the α_r fill $[4h, 5h-1]$; consecutive β_r leave gaps of exactly 6 indices, and the remaining interval $[12h-6,\ 2h^2-1]$ has even length $2h^2 - 12h + 6 \geq 0$ because $h \geq 7$. So the unused indices form $3(h-1) + (h^2 - 6h + 3) = h(h-3)$ dominoes, as required. □

Proof of Lemma 3.12(2). Start with a 7×7 magic square and border it successively to the orders $9, 11, \dots, n$ by the following step, which turns a semi-magic square T of order $n-2$ into one of order $n = 2h+1$ ($h \geq 4$) whose top and bottom rows are complementary on the inner columns. Let $N = n^2$, $c = (N+1)/2$, $A = 4h$, $B = 4h-2$. The inner block is $T + (2n-2)$, which uses the values $2n-1, \dots, N-2n+2$; each inner row and column then sums to $M(n-2) + (n-2)(2n-2) = (n-2)(N+1)/2$. The border uses the $4h = 2n-2$ complementary pairs $\{p, N+1-p\}$, $1 \leq p \leq 4h$. Put $s_{00} = N+1-A$, $s_{n-1,n-1} = A$, $s_{0,n-1} = B$, $s_{n-1,0} = N+1-B$. Let $\mathcal{P} = [2h, 4h] \setminus \{A, B\}$ and $\mathcal{Q} = [1, 2h-1]$, both of size $2h-1$, so $\sum \mathcal{P} - \sum \mathcal{Q} = 4h^2 - 4h + 2$. Choose an h-subset T_+ of $[2h, 4h-3] \subset \mathcal{P}$ and an $(h-1)$-subset T_- of $\mathcal{Q}$ with $\sum T_+ - \sum T_- = c - A + B = 2h^2 + 2h - 1$. This is possible: the subsets of a given size of an interval of integers have as sums all integers between the smallest and the largest such sum (raise one element by 1 at a time), so $\sum T_+ - \sum T_-$ takes every integer in $[h^2+h,\ 3h^2-2h]$, which contains $2h^2+2h-1$ for $h \geq 4$. The inner top row consists of the values $N+1-p$ ($p \in T_+$) and p ($p \in T_-$); the inner bottom row of their complements in the same columns; the inner left column of $N+1-p$ ($p \in \mathcal{P} \setminus T_+$) and p ($p \in \mathcal{Q} \setminus T_-$); and the inner right column of their complements in the same rows. The top row sums to $(h+1)(N+1) - A + B - \sum T_+ + \sum T_- = (2h+1)c = M(n)$, and the left column to $(h+1)(N+1) - A - B - \sum(\mathcal{P} \setminus T_+) + \sum(\mathcal{Q} \setminus T_-) = M(n)$, because $\sum(\mathcal{P} \setminus T_+) - \sum(\mathcal{Q} \setminus T_-) = 4h^2 - 4h + 2 - (2h^2 + 2h - 1) = c - A - B$. The bottom row and right column then sum to $n(N+1) - M(n) = M(n)$, each inner row and column to $N + 1 + (n-2)(N+1)/2 = M(n)$,

and every value $1,\dots,N$ occurs once. Bordering shifts the inner square by a constant ω, and $(u+\omega)+(u'+\omega)=N+1$ whenever $u+u'$ equals the previous $N+1$, so earlier complementary relations persist; this gives the complementary rows $\ell, n-1-\ell$ on the columns $\ell+1,\dots,n-2-\ell$ for $0\le\ell\le(n-9)/2$. □

C Derivation of the variance formulas

Joint moments. Let b be a vector with $\sum_v b_v=0$, $p_r=\sum_v b_v^r$, and z a uniformly random permutation of b. For distinct cells, $\mathbb{E}[z_a^4]=p_4/N$, and with $(N)_k=N(N-1)\cdots(N-k+1)$,

$$\mathbb{E}[z_a^3 z_b]=\frac{-p_4}{(N)_2},\quad \mathbb{E}[z_a^2 z_b^2]=\frac{p_2^2-p_4}{(N)_2},\quad \mathbb{E}[z_a^2 z_b z_c]=\frac{2p_4-p_2^2}{(N)_3},\quad \mathbb{E}[z_a z_b z_c z_d]=\frac{3p_2^2-6p_4}{(N)_4},$$

by Möbius inversion of the power sums over set partitions. Write m_ν for $\mathbb{E}[z_a z_b z_c z_d]$ when the equality pattern of (a,b,c,d) is exactly the partition ν of the positions $\{1,2,3,4\}$, and $m_\lambda=m_\nu$ for any ν of block type λ (e.g. m_{211} for $12|3|4$); the displayed moments are m_4, m_{31}, m_{22}, m_{211} and m_{1111}.

Proof of Proposition 4.3. The mean: $\mathbb{E}[z^{\mathsf T}Az]=\frac{p_2}{N}\operatorname{tr}A-\frac{p_2}{N(N-1)}\sum_{a\neq b}A_{ab}$ and $\sum_{a\neq b}A_{ab}=-\operatorname{tr}A$. For the second moment let $S^{=}(\nu)$ be the sum of $A_{ab}A_{cd}$ over the tuples with pattern exactly ν and $S^{\ge}(\nu')=\sum_{\nu\ge\nu'}S^{=}(\nu)$ the sum over tuples satisfying at least the equalities of ν'; $S^{\ge}(\nu')$ is an unrestricted sum. Möbius inversion gives $\mathbb{E}[(z^{\mathsf T}Az)^2]=\sum_\nu m_\nu S^{=}(\nu)=\sum_{\nu'}\theta(\nu')S^{\ge}(\nu')$ with $\theta(\nu')=\sum_{\nu\le\nu'}\mu(\nu,\nu')m_\nu$. If ν' has a singleton block, the corresponding index occurs in a single factor of $A_{ab}A_{cd}$ and summing over it gives a row sum, so $S^{\ge}(\nu')=0$. The remaining partitions are $12|34$, $13|24$, $14|23$ and 1234, with $S^{\ge}$ equal to $(\operatorname{tr}A)^2$, $\operatorname{tr}A^2$, $\operatorname{tr}A^2$, $\sum_a A_{aa}^2$, and

$$\theta(12|34)=\theta(13|24)=\theta(14|23)=m_{22}-2m_{211}+m_{1111},$$
$$\theta(1234)=m_4-4m_{31}-3m_{22}+12m_{211}-6m_{1111}.$$

For the centred values $1,\dots,N$, $p_2=N(N^2-1)/12$ and $p_4=N(N^2-1)(3N^2-7)/240$, and these coefficients simplify to $N(N^2-1)(5N+6)/720$ and $-N^2(N+1)^2/120$; subtracting $\left(\frac{N(N+1)}{12}\operatorname{tr}A\right)^2$ gives (16).

The invariants. With $\Gamma_{\ell\ell'}=\langle\tilde\varphi_\ell,\tilde\varphi_{\ell'}\rangle$ and $h_c=A_{cc}=\sum_\ell a_{\ell,c}^2$,

$$\operatorname{tr}A=\frac{1}{N}\operatorname{tr}\Gamma,\qquad \operatorname{tr}A^2=\frac{1}{N^2}\sum_{\ell,\ell'}\Gamma_{\ell\ell'}^2,\qquad \sum_a A_{aa}^2=\sum_c h_c^2.$$

For a line, $(\mathbf{1}_L(c)-\frac{1}{n})^2=\frac{1}{n^2}+(1-\frac{2}{n})\mathbf{1}_L(c)$. Hence $N^2h_c=x_c^2+y_c^2+\frac{2}{n^2}+(1-\frac{2}{n})(d_{\mathrm m}+d_{\mathrm a})(c)$ for E_4 and $N^2h_c=\frac{2n+2}{n^2}+(1-\frac{2}{n})(2+d_{\mathrm m}(c)+d_{\mathrm a}(c))$ for E_{full}, where $d_{\mathrm m},d_{\mathrm a}$ are the diagonal indicators. The entries of Γ are given in the proof of Proposition 4.1. The sums needed are $\sum_c x_c^2=\frac{n^2(n^2-1)}{12}$, $\sum_c x_c^4=\frac{n^2(n^2-1)(3n^2-7)}{240}$, $\sum_c x_c^2y_c^2=\left(\frac{n(n^2-1)}{12}\right)^2$, $\sum_{c\in D}x_c^2=\sum_{c\in D}y_c^2=\frac{n(n^2-1)}{12}$ for either diagonal D, $\sum_c(d_{\mathrm m}+d_{\mathrm a})=2n$ and $\sum_c(d_{\mathrm m}+d_{\mathrm a})^2=2n+2\,|D_{\mathrm m}\cap D_{\mathrm a}|$, where $|D_{\mathrm m}\cap D_{\mathrm a}|$ is 1 for odd and 0 for even n. Collecting terms gives the following invariants; where two values are given, the first is for odd and the second for even n. For E_4:

$$\begin{aligned}
\operatorname{tr}A&=\frac{(n-1)(n^3+n^2+12)}{6n^4},\\
\operatorname{tr}A^2&=\frac{(n-1)^2(n^6+2n^5+n^4+144)}{72n^8},\qquad \frac{n^8-2n^6+n^4+144n^2-288n+288}{72n^8},\\
\sum_a A_{aa}^2&=\frac{(n-1)(7n^7+7n^6-13n^5+107n^4+240n^2-720n+720)}{180n^{10}},\\
&\qquad\frac{7n^8-20n^6+120n^5-107n^4+240n^3-1320n^2+2880n-2160}{180n^{10}}.
\end{aligned}\tag{18}$$

For E_{full}:

$$\begin{aligned} \operatorname{tr} A &= \frac{2(n^2-1)}{n^4}, \qquad \operatorname{tr} A^2 = \frac{2(n-1)(n^2+n-1)}{n^8}, \quad \frac{2(n^3-2n+2)}{n^8}, \\ \sum_a A_{aa}^2 &= \frac{2(n-1)(2n-1)(n^2+2n-2)}{n^{10}}, \quad \frac{2(2n^4+n^3-10n^2+12n-6)}{n^{10}}. \end{aligned} \tag{19}$$

D A witness for $k^*(5) = 4$

The arrangement

$$\begin{pmatrix} 1 & 22 & 11 & 24 & 7 \\ 23 & 3 & 12 & 2 & 25 \\ 18 & 6 & 19 & 17 & 5 \\ 4 & 21 & 14 & 10 & 16 \\ 20 & 9 & 15 & 8 & 13 \end{pmatrix}$$

has all row sums $M(5) = 65$ and column sums $66, 61, 71, 61, 66$, i.e. $\gamma = (1, -4, 6, -4, 1)$ and $P_\gamma = (1-t)^4$; it satisfies the degree-3 conditions without being semi-magic. The witnesses for $6 \le n \le 40$ are in `anc/data_witnesses_variances.json` ($n \le 13$) and `anc/kstar_witnesses.json` ($14 \le n \le 40$).